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PRISM a diagnostic framework for parametric recall degradation in quantized large language models

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Abstract

Post-training quantization (PTQ) is the standard method for deploying large language models (LLMs), though it is widely assumed to degrade model quality uniformly. Although, this assumption masks how quantization uniquely affects different mechanisms, particularly the balance between a model's internal parametric memory and its ability to utilize external context. Existing benchmarks primarily measure overall accuracy, failing to isolate these specific sources of knowledge degradation when bit-rates are reduced. To address this, we introduce PRISM (Precision-Resolved Inspection of Stored Memory), an efficient diagnostic framework that separates and measures these distinct capabilities on a single 8GB consumer GPU. PRISM isolates these effects by leveraging counterfactual knowledge-conflict probes, component-wise mixed-precision quantization, and leak-free comprehension probes. We evaluate this framework across multiple model families (Qwen2.5, Llama-3.2, Gemma-2) and diverse quantization schemes (round-to-nearest, NF4, GGUF, GPTQ). Our results demonstrate that open-ended parametric recall collapses under low-bit PTQ while verbatim copy-from-context remains robust; furthermore, component-wise probing reveals that MLP/FFN quantization drives this recall collapse far more than attention layers. Ultimately, these insights reveal that while closed-book benchmarks overstate quantization harm for simple retrieval tasks, complex multi-hop reasoning remains vulnerable, providing crucial nuance for deploying compressed LLMs.

Clinical trial

Not applicable.

Keywords Large language models, Quantization, Model compression, Retrieval-augmented generation, Mechanistic interpretability, Knowledge conflict

1 Introduction

Post-training quantization (PTQ) is the default tool for fitting large language models (LLMs) onto commodity hardware. By reducing weights from 16-bit floating point to 8, 4, or even fewer bits, methods such as GPTQ [1] and activation-aware schemes make once-prohibitive models runnable on a single consumer GPU. The tacit assumption behind reporting a single accuracy number per bit-width is that quantization degrades a

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model *uniformly*: lower precision is taken to mean a model that is uniformly worse, trading a few points of quality for a smaller memory footprint. Scaling laws for low-bit quantization [2, 3] reinforce this view by summarizing compression as a scalar loss against full precision.

We argue this framing is incomplete, because an LLM answers a question from two distinct sources. One is its *parametric memory*, the facts absorbed into the weights during pretraining, and the other is *in-context evidence* supplied at inference time. These two abilities are not interchangeable. Retrieval-augmented generation (RAG) [4–9], tool use, and agentic pipelines are all built on the second: the model is expected to read evidence placed in its context and act on it, often *in spite of* what its weights "believe." A deployment number that mixes both sources can therefore mislead exactly the practitioners who quantize most aggressively. Prior work hints that the two abilities are not equally fragile: fact recall deteriorates before in-context learning under parameter downscaling [10], and quantization measurably perturbs factual recall [11], yet neither asks how compression resolves a *conflict* between memory and context, which is the regime that governs RAG. We close this gap with PRISM (Precision-Resolved Inspection of Stored Memory), a diagnostic framework that measures, separately, how quantization affects parametric memory and the use of in-context evidence, combining controlled counterfactual knowledge-conflict probes, component-wise mixed-precision quantization, and a leak-free comprehension probe, run entirely on a single 8 GB consumer GPU.

Our central finding is that low-bit PTQ erodes capabilities by *source*, not uniformly. Across both Qwen2.5 models (3B and 7B), open-ended *parametric recall* degrades far faster than the model's ability to use *in-context* evidence: on Qwen2.5-7B with `llama.cpp` k-quants, pushing precision from 6-bit down to 2-bit lowers closed-book accuracy from 1.00 to 0.88 while the rate at which the model follows a counterfactual passage that contradicts its memory does not fall, and the same direction holds on Qwen2.5-3B. The cleanest way to see that context use is preserved is a *leak-free* multi-hop comprehension probe over fictional entities, whose closed-book accuracy is 0.00 by construction: there, genuine compose-from-context comprehension survives to 4-bit (0.74) while parametric recall has already collapsed. Copy-from-context is thus the most robust capability under PTQ, recall the most fragile, and multi-hop comprehension intermediate, a fragmentation by source rather than a uniform decline, and one that holds across quantizers (round-to-nearest, NF4, GGUF *k*-quants, and calibrated GPTQ) and across all three model families we test.

To probe *where* in the network this fragility lives, PRISM uses component-wise mixed-precision quantization, quantizing *only* the MLP/FFN sublayers, or *only* attention, while holding the rest at full precision. Quantizing the MLP erodes recall far more than quantizing attention (closed-book accuracy 1.00 → 0.16 at 3-bit on Qwen2.5-3B for MLP-only, versus 1.00 → 0.86 for attention-only), this component-level result *replicates across three model families* (on Llama-3.2-3B and Gemma-2-2B the same component-wise quantization erodes recall under MLP quantization, ΔCB = 0.38 and 0.36 respectively, but spares it under attention quantization, ΔCB = 0.03 and 0.02, with attention precision-robust everywhere), and it survives a change of quantizer (under a calibrated GPTQ quantizer the same contrast persists, ΔCB = 0.62 for MLP versus 0.08 for attention). We are deliberately careful not to over-read this component-level result, for two reasons. First, it is *weight-mass-confounded*: the MLP holds about 88% of the model's

parameters, so quantizing it is also the larger perturbation, and at *equal per-parameter* sensitivity MLP and attention are comparable. Second, the sharper, mass-controlled localization we observe under naive round-to-nearest, where at equal perturbed mass ($\sim$340 M parameters) quantizing the *early* MLP layers erases more recall (ΔCB = 0.667) than all of attention (0.135) or the middle MLP (0.010), is *not* quantizer-general: a calibrated GPTQ quantizer eliminates the equal-mass early-MLP effect entirely (early-five-MLP ΔCB = 0.000), and at equal mass attention then erases *more*. The *depth* of the effect is, by contrast, model- and quantizer-specific: the early-layer concentration is a naive-round-to-nearest property of Qwen, absent on Gemma-2-2B (where the early-five MLP layers barely matter, ΔCB = 0.019) and absent under a calibrated GPTQ quantizer even on Qwen. We therefore present the early-MLP localization as a round-to-nearest-and-Qwen-specific observation rather than a general per-parameter claim: the cross-family, cross-quantizer claim is the *component*-level one (recall is the most MLP-fragile capability, attention robust), while the early-MLP refinement is a single-quantizer, single-family observation. Because our causal evidence rests on a single instrument, component-wise quantization, we describe the MLP as the *locus of the perturbation-induced recall collapse* rather than where the fact is stored, relating our findings to the view that feed-forward layers act as key–value memories [12] and that factual associations live in MLPs [13], now read through the *precision* axis.

Two honest tempering points are central, not footnotes. First, the erasure is *graded, not surgical*: at the 3-bit precision where recall fails, MLP quantization also raises WikiText-2 perplexity about 4.5× and halves MMLU. Quantization does not erase facts while leaving the model otherwise intact; rather, recall is the *most* fragile capability and copy-from-context the *most* robust, with fluency and multiple-choice reasoning in between. Second, the apparent *rise* in context-following is largely an artifact of *removed competition*: when MLP quantization raises the counterfactual-following rate from 0.60 to 0.82, about 96% of the newly-following items are exactly the items that also lost their closed-book memory; on the memory-intact subset the rate barely moves (0.667 $\rightarrow$ 0.733). We therefore lead the context-ability story with the leak-free comprehension probe rather than with this rate.

The deployment picture is consequently more nuanced than "quantize and lose a few points everywhere." Closed-book benchmarks *overstate* the harm of quantization for the simplest retrieval-augmented use: verbatim single-hop copy-from-context, which much RAG depends on, is the capability that survives, while it is MLP-dependent parametric recall that quantization preferentially destroys. But multi-hop retrieval is *not* spared: a harder 3-hop task (fictional entities, distractors, closed-book 0.00) degrades gracefully to 4-bit (0.81) and then collapses at uniform 3-bit, and MLP quantization wrecks it (0.14) while attention quantization largely preserves it (0.64). Practitioners can safely quantize for simple copy-RAG, but should not assume multi-hop reasoning over retrieved context survives aggressive low-bit PTQ (see Fig. 1).

PRISM is organized around four questions. First, does low-bit PTQ degrade an LLM's capabilities *uniformly*, or does it fragment them by *source*, separating open-ended parametric recall from the use of in-context evidence (RQ1)? Second, *where* in the architecture does any quantization-induced recall loss localize, and is that localization a genuine component effect or an artifact of weight mass and of the particular quantizer (RQ2)? Third, does the recall-versus-context split *generalize* across model families and

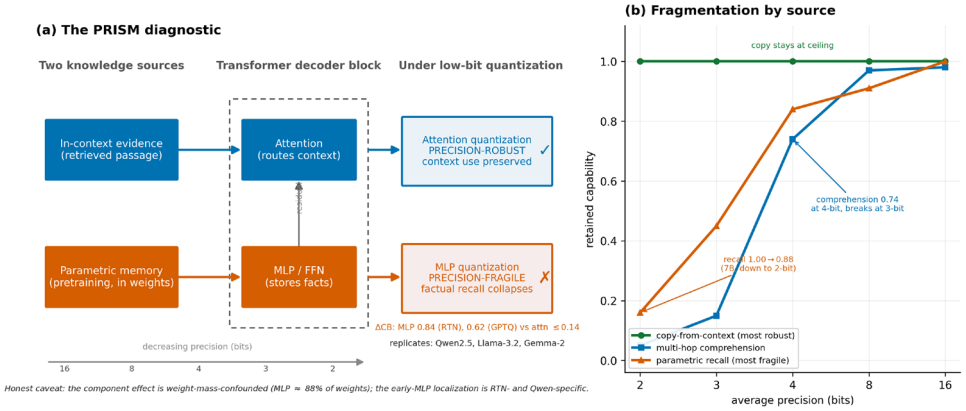


Fig. 1 Overview of PRISM. (**a**) As PTQ precision drops, closed-book recall falls steeply while copy-from-context is preserved. (**b**) Component-wise, quantizing the MLP erodes recall far more than attention (round-to-nearest $\Delta\mathrm{CB} = 0.84$, calibrated GPTQ 0.62)

quantization methods, and to how low a bit-width does it persist (RQ3)? Fourth, what are the *deployment* consequences for retrieval-augmented generation, specifically how single-hop copy-from-context and multi-hop composition fare under aggressive quantization (RQ4)?

Our contributions answer these questions in turn. First, we introduce PRISM, a single-GPU diagnostic framework that separates a quantized model's parametric recall from its use of in-context evidence, and use it to establish the capability fragmentation above. Second, via its component-wise probe we give a component-level mechanistic account of where recall fragility lives, stated with its honest weight-mass and early-MLP caveats. Third, its leak-free comprehension probe and removed-competition analysis provide a measure of context use that no parametric leakage can inflate. Relative to prior work, PRISM is, to our knowledge, the only framework that jointly manipulates *precision*, places parametric memory in *conflict* with in-context evidence, and *causally localizes* the resulting split by component across model families and quantizers; Sect. 2 makes this positioning explicit against the closest prior studies (Table 2), which each address at most a subset of these axes and operate at full precision.

For a practitioner, PRISM is used as a two-probe diagnostic on a candidate quantized checkpoint. Running the closed-book knowledge-conflict probe and the leak-free comprehension probe reads off, *separately*, how much stored parametric recall and how much context-use each bit-width costs, rather than collapsing both into a single accuracy number. A deployment that answers from the model's own knowledge (closed-book question answering) should therefore treat low-bit quantization cautiously, whereas a deployment that supplies facts through retrieval can quantize far more aggressively, provided its retrieval is single-hop copy rather than multi-hop composition. Section 4 distils these into concrete per-regime precision floors.

The remainder of this paper is organized as follows. Section 2 reviews the literature and positions our contribution; Sect. 3 details the methodology and experimental setup, including the models, quantizers, datasets, metrics, and the conflict classifier that make up PRISM; Sect. 4 presents our results and analysis, covering the capability-fragmentation phenomenon, the component-level mechanism and its controls, the precision-allocation corollary, and the multi-hop and single-hop retrieval analysis; Sect. 5 states limitations and threats to validity; and Sect. 6 concludes.

2 Literature review

Table 1 summarizes the prior work most relevant to PRISM across five threads: efficient quantization methods, the measured impact of quantization on capability, the asymmetry between memorized and in-context knowledge, where factual knowledge is localized in a transformer, and knowledge conflict and RAG. We expand on each below.

Post-training quantization (PTQ) is the standard route to deploying LLMs under tight memory budgets [22]. Weight-only methods (GPTQ [1], AWQ [14]), activation-smoothing schemes [15, 23], and quantization-aware finetuning [16] push usable inference toward 4-bit, while more aggressive codebook and sparse-outlier techniques [24–27] reach 2-bit. A parallel line characterizes *what* compression costs: precision-aware scaling laws [2, 28], dependence on training extent [3], and broad capability evaluations [17, 29–31]; most directly [11], shows factual recall degrades with precision. These works establish that quantization erodes stored knowledge but treat closed-book competence

Table 1 Representative prior work by thread: per-row focus and method, key finding, and gap relative to PRISM (cf. The positioning table, table 2)

Author & Year	Focus & method	Key finding	Gap relative to this work
Efficient quantization methods			
Frantar et al. (2022) [1]	Weight-only PTQ; GPTQ	One-shot 3–4-bit quantization of large models at near-lossless perplexity	Evaluates fluency, not factual recall
Lin et al. (2024) [14]	Activation-aware PTQ; AWQ	Protects salient weight channels for strong 4-bit inference	No parametric-memory or context analysis
Xiao et al. (2023) [15]	Weight+activation PTQ; SmoothQuant	Migrates activation outliers into weights for near-lossless INT8	High-bit only; retention not studied
Dettmers et al. (2023) [16]	4-bit finetuning; NF4/QLoRA	NF4 with double quantization enables near-FP16 4-bit finetuning	Efficiency-focused; memory effects not isolated
Impact of quantization on capability			
Liu et al. (2025) [2]	Low-bit scaling laws	Charts accuracy as a Pareto function of bit-width	Aggregate loss; no memory-vs-context split
Ouyang et al. (2024) [3]	Quantization vs. training extent	Low-bit loss grows with how thoroughly a model was trained	Perplexity-level; recall not measured
Li et al. (2024) [17]	Broad eval of quantized LLMs	Benchmarks how reasoning and emergent abilities survive	Aggregate competence; no conflict setting
Wang et al. (2025) [11]	Quantization & factual recall	Factual knowledge degrades as precision drops	Closed-book only; no conflict or localization
Memorized vs. in-context knowledge			
Jin et al. (2023) [10]	Capacity reduction (pruning, scaling)	Recall deteriorates before in-context learning	Pruning not quantization; no conflict; no mechanism
Where knowledge lives in a transformer			
Meng et al. (2022) [13]	Locating & editing facts; ROME	Facts localize to mid-layer MLP and are editable	Full precision; quantization not studied
Geva et al. (2021) [12]	Mechanism of FFN layers	Feed-forward layers act as key–value memories	Architectural axis only; precision fixed
Dai et al. (2022) [18]	Knowledge neurons	FFN neurons tie to specific relational facts	Full precision; no compression
Knowledge conflict and RAG			
Longpre et al. (2021) [19]	Knowledge conflict; entity substitution	Reveals when models follow memory vs. context	Full precision; precision not varied
Xie et al. (2024) [20]	Memory–context arbitration	Models swing between trusting memory and context	Full precision; no compression
Neeman et al. (2023) [21]	Disentangling sources; DisentQA	Separates parametric from contextual answers	Full precision; no quantization
Lewis et al. (2020) [4]	Retrieval-augmented generation	Retrieval augments parametric memory with context	No compression; context assumed trusted

in isolation: none study how quantization resolves a memory-versus-context *conflict*, allocate precision across MLP and attention, or push to $\leq$ 2-bit on this question.

Our precision-allocation corollary (Sect. 4) is also distinct from existing mixed-precision quantization in *what* drives the allocation. Methods such as AWQ [14] and SmoothQuant [15] preserve model quality through importance metrics, activation-outlier statistics, or activation-aware transformations, allocating or protecting precision according to conventional quantization heuristics. Our approach differs by grounding precision allocation in *experimentally observed component-level sensitivity* to quantization. Specifically, we find that MLP sublayers exhibit substantially greater degradation in parametric recall than attention under the evaluated settings, and we allocate precision on the basis of that measured behavior rather than a weight- or activation-importance proxy. The contribution is therefore not a new mixed-precision heuristic but a demonstration that where quantization breaks a model, measured directly by capability, is itself an actionable allocation signal.

Closest to our phenomenon [10], reports that as models are down-scaled, fact recall deteriorates before in-context learning, an asymmetry we are consistent with but differ from on every operative axis: it reduces capacity by *pruning* rather than quantizing a fixed architecture, uses *supporting* rather than counterfactual context, offers no component-level mechanism, and does not vary precision. PRISM instead holds the architecture fixed, varies only bit-width, places memory and context in opposition, and causally localizes the split.

A substantial knowledge-conflict literature [32] studies how models arbitrate between parametric beliefs and in-context evidence, via entity-substitution [19], faithfulness stress-tests [33], counterfactual QA [21], and analyses of when models trust memory versus context [20, 34, 35], alongside behavioral fixes through prompting [36] or decoding [37]. We adopt this conflict framing and its negation-aware spirit, but every one of these studies operates at full precision: PRISM's contribution is to introduce precision as the manipulated factor and show it moves the conflict outcome in a mechanistically interpretable direction.

A mechanistic-interpretability literature separates the two sublayers: attention moves and copies information between positions [38–40], while feed-forward layers store and promote content, behaving as key–value memories [12, 41] that recall factual associations in mid-to-late MLP layers [42, 43] and can be located and edited module-by-module [13, 18, 44], typically via activation patching and causal tracing [45]. These pin the MLP as the locus of parametric memory along the *architectural* axis (the competence MMLU [46] measures). PRISM extends this to the orthogonal *precision* axis, showing component-wise that quantizing only the MLP collapses closed-book memory while sparing copy-from-context, replicating across three families. We engage one counterpoint [47]: shows causal-tracing localization need not predict where editing succeeds, which is precisely why our central evidence is a direct weight-level *intervention*, with activation patching a confounded secondary signal.

To our knowledge PRISM is the first framework to (a) study PTQ's effect on knowledge-*conflict* resolution, (b) *causally* localize the knowledge-versus-context split via component-wise MLP-vs-attention quantization, and (c) probe this trade-off to 2-bit. Table 2 positions this work against the closest threads; no prior line combines a

Table 2 Positioning vs. The closest prior work: PRISM is the only row that jointly satisfies compression, conflict, component localization, varied precision, ≤2-bit, cross-family, and RAG framing. ✓=yes, ×=no, (∼)=partial, "–"=N/A

Work	Compres-sion	Context	Com-ponent localiz.?	Precision varied?	≤2-bit?	Cross-family?	RAG fram-ing?
Jin et al. [10]	prune	support	×	×	×	×	×
Wang et al. [11]	quant	none	×	✓	×	×	×
Longpre et al. [19, 33, 34]	none	conflict	×	×	–	–	(∼)
Meng et al. [12, 13, 18]	none	none	✓†	×	×	×	×
PRISM (this work)	quant	conflict	✓	✓	✓	✓	✓

†Localizes facts for *editing* at full precision; we localize the *quantization*-induced memory loss

compression manipulation, a knowledge-*conflict* setting, component *localization*, varied *precision* to 2-bit, *cross-family* replication, and an explicit *RAG* framing.

3 Methodology and experimental setup

PRISM is designed to run on commodity hardware. All experiments run on a single consumer GPU, an 8 GB RTX 5060, in an inference-only regime: we never fine-tune or calibrate on the probe data, so every result is reproducible on commodity hardware without a training cluster. This is a deliberate design choice, it makes the full pipeline (quantize, probe, score) replayable end-to-end on one workstation, but it also bounds our scope: the 8 GB budget restricts us to models of ≤ 7B parameters and, for the component-wise quantization of Sect. 4, to a single 3B base that fits in full precision alongside its quantized variant. We do not study 13–70B models. The computational overhead of PRISM itself is modest and dominated by a handful of greedy-decode passes over small probe sets: on the 8 GB GPU a single (model, quantization) configuration completes in roughly five to fifteen minutes, and the full evaluation suite in a few GPU-hours. Quantization is one-shot per configuration, seconds for round-to-nearest and NF4, a few minutes for GPTQ on its 256 WikiText-2 calibration samples. No step requires gradient computation or a training cluster.

We use four instruction-tuned models: `Qwen2.5-3B-Instruct` and `Qwen2.5-7B-Instruct` [48], `Llama-3.2-3B-Instruct` [49], and `Gemma-2-2B-it` [50]. Qwen2.5-3B is our primary model for the phenomenon, the causal mechanism, and the equal-perturbed-mass control; Qwen2.5-7B provides the cleanest bit-width sweep. Llama-3.2-3B and Gemma-2-2B are the two additional families on which we test the cross-family generality of the component-level memory→MLP localization. Each additional family fits in full precision alongside its quantized variant within the 8 GB budget, enabling the same component-wise quantization as on the 3B primary.

We span four post-training quantization (PTQ) toolchains so that conclusions are not an artifact of one implementation. bitsandbytes supplies INT8 [23] and the 4-bit NormalFloat (NF4) scheme of QLoRA [16] (double-quantized, `bf16` compute). torchao supplies round-to-nearest (RTN) weight-only quantization (`IntxWeightOnlyConfig`), applied per-group with group size 32 and *asymmetric* zero-points, at 8/4/3/2 bits. GPTQ [1] (via `gptqmodel`, group size 128, calibrated on 256 WikiText-2 samples) provides a strong, calibrated weight-only baseline, and, via its per-module `dynamic` configuration that skips all non-target modules, a *second, calibrated* component-wise quantizer that

lets us test whether the component-level localization of Sect. 4 is an artifact of the naive round-to-nearest scheme. Module targeting is verified by counting the quantized modules (108 MLP / 144 attention projections; 15 early-/15 mid-MLP layer bands). GGUF k -quants via `llama.cpp` [51] provide the $Q6/Q4/Q3/Q2$ sweep on the 7B model, with all layers offloaded to GPU. Crucially, torchao and GPTQ expose *component-wise mixed precision*: a per-module name filter (torchao, matching. `mlp.` versus `self_attn`) or a per-module `dynamic` configuration (GPTQ) lets us quantize *only* the MLP/FFN sublayers or *only* the attention projections, at independently chosen bit-widths. This is the lever for the causal localization in Sect. 4 and the precision allocation in Sect. 4. The language-model head (`lm_head`) is kept in FP16 across all configurations so it cancels out of every comparison. Table 3 collects the full configuration, including software versions, for reproducibility.

We use three datasets: an in-house knowledge-conflict set (FACTS_V2), a leak-free multi-hop comprehension probe (COMPREHENSION), and the standard COUNTERFACT benchmark for external validity. The first two are constructed so that the model's score reflects a genuine memory-versus-context decision rather than surface copying.

The first dataset, FACTS_V2, is for knowledge conflict. It contains 114 moderate-prior, *updatable* facts that the full-precision model already knows, drawn from five relations: company headquarters city (32), company founding year (32), company CEO (20), film director (18), and composer nationality (12). Each item pairs a gold fact with a

Table 3 Experimental settings and hyperparameters: inference-only on a single 8 GB GPU, with no fine-tuning or calibration on probe data; weight-only quantization keeps `lm_head` in FP16 throughout

Component	Setting
Hardware	single NVIDIA RTX 5060 Laptop GPU (8 GB), inference-only
Models	`Qwen/Qwen2.5-3B-Instruct` (primary; mechanism, mass control)
	`Qwen/Qwen2.5-7B-Instruct` (bit-width sweep)
	`unsloth/Llama-3.2-3B-Instruct` (cross-family)
	`google/gemma-2-2b-it` (cross-family)
Decoding	greedy (`do_sample=false`, `num_beams=1`),
	`max_new_tokens` = 32 (conflict/copy), **96** (3-hop CoT),
	model chat template, one gen/item
torchao RTN	`IntxWeightOnlyConfig`, group size **32**, asymmetric, per-group;
	component targeting by module-name filter (**`.mlp.`** vs. **`self_attn`**);
	early/mid/late MLP *layer-band* filters for the mass control
Weight-mass control	equal perturbed mass (∼340 M params) at 3-bit:
	early-**5** MLP vs. all-attention vs. mid-**5** MLP;
	cumulative MLP layer prefixes; per-layer and per-100 M sweeps
MMLU	**1995**-item stratified subset; letter-logprob scoring [46]
Perplexity	WikiText-2 test, sliding window **512**
bitsandbytes	INT8; NF4 (**4**-bit, double-quant, `bf16` compute)
GPTQ	group size **128**, **256** WikiText-2 calibration samples,
	component-wise via the `dynamic` config (skip non-target modules);
	cross-quantizer control: MLP-/attn-only and early-/mid-MLP at 3-bit
GGUF k-quants	$Q6_K/Q4_K_M/Q3_K_M/Q2_K$ (`llama.cpp`), all layers on GPU
Bootstrap	paired percentile bootstrap, $B = 2000$ ($B = 10000$ for the mechanism)
Seeds	headline runs: greedy, one generation per item (deterministic)
	generation-variance check: sampled at temperature **0.7**, **3** seeds
	(MLP-3-bit), plus the greedy run
Software	PyTorch **2.11+cu128**; `transformers`, `torchao`,
	`bitsandbytes`, `gptqmodel`, `llama.cpp`

counterfactual obtained by a *within-relation swap* (e.g., replacing one company's HQ city with another company's HQ city), so the counterfactual is type-correct and plausible rather than nonsensical. The counterfactual is then embedded in a fluent, RAG-style passage. The model is queried with the same question under three context conditions, the counterfactual passage (*conflict*), a passage stating the true fact (*support*, for open-book accuracy), and an irrelevant passage, enabling both the conflict-resolution and the retrieval-recovery analyses. A representative conflict item:

> *Question: In which city is the company Google headquartered?*
> *Gold: Mountain View Counterfactual: Dearborn*
> *Conflict passage: "According to a recent corporate filing, Google confirmed that its global headquarters is located in Dearborn. The company has consolidated its primary operations and executive offices in Dearborn."*

Our second probe is a leak-free multi-hop COMPREHENSION set. To rule out the objection that open-book performance is mere copying, we build 160 synthetic two-hop items over *fictional* entities (company → city → nation), each accompanied by two distractor chains. Answering requires *composing* two context sentences (which city a company sits in, and which nation that city sits in); the gold answer never appears verbatim next to the queried entity. Because the entities are invented, closed-book accuracy is 0*by construction*, so any accuracy is attributable purely to in-context composition rather than memory or copying. A representative item:

> *Question: In which nation does Quodax Corporation have its headquarters?*
> *Gold: Hothxan Counterfactual: Jundmar*
> *Support passage: "Quodax Corporation has its headquarters in Eskjiv City. Eskjiv City lies within the nation of Hothxan. Vexmar Holdings has its headquarters in Qanzeph ton. Qanzeph ton lies within the nation of Maripsa. Rowgrim Corporation has its headquarters in Xanwyn City. Xanwyn City lies within the nation of Belkor."*

To probe harder, non-ceiling multi-hop reasoning over retrieved context, rather than single-hop verbatim copy, we additionally build 80 synthetic *3-hop* items over fictional entities with non-adjacent gold facts and distractor chains, scored with full chain-of-thought generation. As with COMPREHENSION, closed-book accuracy is 0 for all configurations by construction, and full-precision open-book accuracy is well below ceiling (0.91), so this set yields a genuine degradation curve rather than a ceiling effect.

To test whether quantization erases facts *selectively* or simply degrades the model broadly, we pair the recall probes with two capability controls: a 1995-item stratified subset of MMLU [46], scored by the log-probability of the answer letter, and WikiText-2 test-set perplexity computed with a sliding window of 512 tokens. These let us state honestly that at the precision where recall fails the model also loses fluency and multiple-choice accuracy (Sect. 4).

As an external benchmark we additionally evaluate COUNTERFACT [13] (the `Neel-Nanda/counterfact-tracing` release) as a standard external set. We report it with the caveat that its cloze format produces substantial garbage (OTHER ≈ 0.25; Sect. 4), so we treat it as supportive-but-noisy evidence rather than a clean headline.

For every dataset the anchor set $\mathcal{A}$ (Sect. 3) restricts scoring to items the full-precision model answers correctly *closed-book*, guaranteeing a real counter-parametric conflict;

for COMPREHENSION, closed-book accuracy is 0 for all configurations by construction, so context is the only possible source of a correct answer. Counterfactual targets are within-relation swaps, never the true value, and the queried entity's answer is never adjacent to its label in the passage.

For metrics, let $\mathcal{A}$ be the *anchor set*: items the full-precision reference answers correctly *closed-book*, so that a genuine counter-parametric conflict exists for every item we score. All headline numbers are computed on $\mathcal{A}$. By construction this fixes the full-precision baseline at CB-Acc = 1.00; we treat this as a stated limitation and verify that *unfiltered* results barely move (e.g., a mechanism CFR of 0.823 anchored versus 0.807 unfiltered). For each item the model's response is classified, negation-aware, into exactly one outcome. We define:

$$\text{CB-Acc} = \frac{1}{|\mathcal{A}|} \sum_{i \in \mathcal{A}} \mathbb{1}[\text{closed-book answer correct}], \tag{1}$$

$$\text{CFR} = \frac{1}{|\mathcal{A}|} \sum_{i \in \mathcal{A}} \mathbb{1}[\text{model asserts the counterfactual}], \tag{2}$$

$$\text{MEM} = p + \text{reject}, \quad \text{OTHER} = 1 - \text{CFR} - \text{MEM}. \tag{3}$$

CB-Acc (closed-book accuracy) measures retained parametric memory with no context. CFR (context-following rate) is negation-aware: it counts only responses that positively *assert* the counterfactual, not ones that negate it, and not ones that restate the prior, so CFR measures a real memory-override decision rather than surface copying. MEM aggregates memory-siding outcomes: p (asserts the prior) plus *reject* (explicitly rejects the counterfactual). OTHER ("garbage") captures responses that assert neither the prior nor the counterfactual; it rises when the model *breaks*, and we always report it. In plain terms: CB-Acc asks "with no context, can the model still answer from what it memorized?"; CFR asks "when the passage contradicts that memory, does the model actually *assert* the new, contradicting fact?", and is negation-aware so that a memory-defending answer phrased around the counterfactual ("Apple is *not* in Round Rock") is not miscounted as context-following; and the leak-free comprehension probe of Sect. 4 asks "can the model answer a question whose answer exists only by *chaining* two supplied facts about invented entities?", so that, since the entities cannot have been memorized, any correct answer must come from reading context rather than from recall or verbatim copying. Any configuration with OTHER > 0.2 is a *model-breakage regime*: its CFR is denominator-diluted and unreliable, so we flag such configurations rather than headlining them.

For the retrieval conditions of Sect. 4 we report open-book / support accuracy: accuracy when the gold fact is supplied verbatim in context.

To state that "memory falls faster than context" precisely, we compare drops from full precision: $\Delta\text{CB} = \text{CB-Acc}_{\text{FP}} - \text{CB-Acc}_q$ and $\Delta\text{CFR} = \text{CFR}_{\text{FP}} - \text{CFR}_q$ at bit-width q. Memory erodes faster than context-following when $\Delta\text{CB} > \Delta\text{CFR}$. We use this single sign convention (a larger drop $\Rightarrow$ larger Δ) throughout. For the retrieval-recovery analysis we report *recovery*$= (\text{OB} - \text{CB})/(1 - \text{CB})$, the fraction of the closed-book gap closed once the gold fact is supplied. Table 4 summarizes all symbols and their sign conventions.

Table 4 Notation and metrics, with the direction a healthier quantized model moves each quantity; all headline metrics are computed on the anchor set $\mathcal{A}$

Symbol	Meaning	Direction
$\mathcal{A}$	anchor set: items FP16 answers correctly closed-book	—
CB-Acc	closed-book accuracy = retained parametric memory	higher = more memory
CFR	context-following rate: asserts the counterfactual (negation-aware)	higher = more override
MEM	memory-siding = p (asserts prior) + *reject* (denies counterfactual)	higher = more memory
OTHER	"garbage": asserts neither prior nor counterfactual	lower = more coherent
ΔCB	drop in CB-Acc from FP16: $\mathrm{CB}_{\mathrm{FP}} - \mathrm{CB}_q$	larger = more erosion
ΔCFR	drop in CFR from FP16: $\mathrm{CFR}_{\mathrm{FP}} - \mathrm{CFR}_q$	larger = more erosion
recovery	$(\mathrm{OB} - \mathrm{CB})/(1 - \mathrm{CB})$: closed-book gap closed by context	higher = better RAG
Regimes		
functional	$\mathrm{OTHER} < 0.2$: model coherent; CFR is trustworthy	target regime
model-breakage	$\mathrm{OTHER} > 0.2$: CFR denominator-diluted; flag, do not headline	avoid

Each conflict response is mapped to exactly one outcome by a deterministic, *negation-aware* rule classifier operating on normalized text (lowercased, punctuation- and article-stripped, whitespace-collapsed). Let the answer string contain the gold value g or the counterfactual c as a whole-token span; a span is *negated* if it follows a negation cue (`not/no/never/` …) within the same clause. Writing $c^+ = [c\text{present} \wedge \neg\text{negated}]$, $p^+ = [g\text{present} \wedge \neg\text{negated}]$, and $\text{rej} = [c\text{negated}]$, the label is assigned in order:

1. c (context-following) if $c^+ \wedge \neg p^+$, asserts the counterfactual and not the prior;
2. p (prior) if $p^+ \wedge \neg c^+$, asserts the prior;
3. reject if rej, explicitly denies the counterfactual (sides with memory);
4. otherwise other (garbage).

CFR counts only label c; memory-siding MEM = p + reject; OTHER is the rest. The negation guard is what separates genuine override ("Apple is headquartered in Round Rock") from memory-defense phrased around the counterfactual ("Apple is *not* headquartered in Round Rock").

We validated the rule classifier against human gold labels from *two independent annotators*, on two samples. On a 60-item sample, a second annotator (independent of the first, blind to the rule's output) hand-assigned the three-way outcome (context-following / memory-siding / garbage); the rule agreed with this annotator on all 60 items, Cohen's $\kappa = 1.00$. On an earlier 46-item sample from `mech__fp16` and `mech__mlp3` (a deterministic stride over item ids, spanning all five relations and both a clean and a degraded model), the first author hand-assigned a gold outcome by reading each (gold, counterfactual, answer) triple, blind to the rule; the rule agreed on 45/46 answers (97.8%), Cohen's $\kappa = 0.95$ (chance agreement $p_e = 0.58$). The single disagreement was a garbled 3-bit MLP-only generation ("… not explicitly stated where Snap is headquartered in Arlington. However, it can be inferred that the company is headquartered there …") where the negation cue misfires on an answer that ultimately affirms the counterfactual, an inherently ambiguous, model-breakage-regime case. This corroborates an earlier 15/15 hand-check on a separate audit sample. We report this agreement honestly: the knowledge-conflict slice consists of short, templated answers, so it is an *easy* labeling task on which near-perfect agreement is expected; agreement would be lower on noisier, free-form slices. The validation script and both annotators' labels are released with the code.

All headline generations use greedy decoding for determinism (`do_sample=false`; `max_new_tokens` = 32 for the conflict and copy probes, 96 for the chain-of-thought scoring of the 3-hop set). To confirm that the recall-collapse headline is not an artifact of a single greedy decode, we additionally run a *generation-variance* check on the round-to-nearest MLP-3-bit configuration: three sampled seeds at temperature 0.7 alongside the greedy run. We select this breakage-adjacent configuration deliberately, rather than a lower-variance or randomly chosen cell, because it represents the highest-variance regime in our sweep; demonstrating stability at this most stringent point implies robustness across the lower-variance settings elsewhere. Closed-book accuracy is stable across seeds (range $[0.146, 0.156]$, standard deviation 0.005), confirming the recall collapse is seed-robust; the context-following rate varies more at this breakage-adjacent cell (range $[0.781, 0.854]$, standard deviation 0.027), so we report sampling standard deviations where the model is near breakage. Confidence intervals are 95% percentile bootstrap intervals over items ($B = 2000$; $B = 10000$ for the mechanism); for component contrasts (Sect. 4) we use the *paired* bootstrap over the shared anchor set.

Two supporting analyses are deferred to appendices. First, an *activation-patching* study (injecting full-precision component outputs into the 3-bit model) is reported as a methodological note: because splicing FP16 activations into an otherwise quantized residual stream is a distribution mismatch, the restoration is non-selective and corroborates the weight-level component-wise quantization only weakly; we therefore treat the direct quantization intervention as the cleaner causal tool. Second, for every anchored headline we provide the corresponding *unfiltered* (full-set) tables, which show the same memory→MLP pattern and confirm that anchoring does not inflate the effect.

4 Results and analysis

We organize our empirical findings as follows. We first establish the headline phenomenon, that low-bit quantization fragments capabilities by source, then give a component-level causal account of where the recall fragility lives, with its weight-mass and cross-quantizer controls. We draw out a precision-allocation corollary for retrieval-augmented deployment, and close with the retrieval analysis that bounds the deployment claim from both the single-hop and multi-hop sides.

Our central claim is empirical and falsifiable, and it is a claim about *which* capabilities low-bit post-training quantization (PTQ) costs. PTQ does not degrade a model uniformly: it fragments its capabilities by their *source*. Open-ended *parametric recall*, answering from stored knowledge with no context, is the *most fragile* capability; verbatim *copy-from-context* is the *most robust*; and multi-hop comprehension sits in between. This ordering is the headline result of the paper, and (unlike the component-level localization of Sect. 4) it is *quantizer- and family-general*: it holds across two quantization back-end families on Qwen, down to 2-bit on a 7B model, and reappears on a standard external benchmark. We measure parametric recall with closed-book accuracy (CB-Acc) and the conflict outcome with the context-following rate (CFR), both defined in Sect. 3, and we always report the garbage rate (OTHER) so that any change is attributable to a behavioral shift rather than to model breakage. Throughout, we restrict headline claims to the *functional regime* (OTHER < 0.2).

As bit-width falls, CB-Acc declines while the most robust capability is the model's ability to copy from supplied context. The measured CFR on conflict items also rises, but

we show this is *removed parametric competition*, not improved context use; the robust context capability is copy-from-context, and our primary positive evidence on context ability is the leak-free comprehension probe of Sect. 4.

Before the detailed evidence, Table 5 consolidates the paper's headline results: one row per finding, each with its anchor size and a one-line claim. The remainder of this section substantiates these rows in turn. All rates carry 95% Wilson confidence intervals in the per-result tables.

Table 7 (top block) reports Qwen2.5-3B-Instruct [48] on the facts_v2 set (moderate-prior, updatable facts; anchor n=96) under five precisions and two quantization back-ends. Within the single TORCHAO RTN sweep, closed-book memory declines smoothly as precision falls, from FP16 (1.00) through 8-bit (0.91) to 4-bit (0.84), while context-following *rises* from 0.60 to 0.74. We restrict the monotone-decline statement to this within-back-end sweep: across *different* back-ends the closed-book column is not monotone (e.g. BNB/INT8 reads 0.97 at 8-bit versus TORCHAO's 0.91), as one expects when comparing distinct quantization implementations. The garbage rate never exceeds 0.05: this is a clean functional regime in which the model is not breaking but shifting *which* source of evidence it trusts. The two 8-bit and two 4-bit back-ends (BNB/INT8 [23], TORCHAO, NF4 [16]) all show the same qualitative direction (memory down, context up), indicating the effect is a property of low-bit quantization rather than of one implementation.

The rising context-following column must be read with care, and we state plainly what it does and does not mean: *context-following did not improve; the parametric competitor was removed*. CFR is measured on knowledge-conflict items where a stored prior competes with the supplied counterfactual, so when quantization erases the prior, the competition vanishes and a flip toward the context is recorded, without the model having become any better at reading context. We make this decoupling quantitative on the component-wise mechanism configuration (Sect. 4), where the effect is largest: MLP-only 3-bit quantization moves CFR from 0.604 to 0.823, an apparent +0.219 rise, but 96% of the newly context-following items (AF = 0.962; 25 of 26 flips) are precisely the items that *also* lost their closed-book memory. On the memory-intact subset, items whose prior survived quantization, CFR barely moves (0.667 → 0.733). The same attribution holds for whole-model 3-bit quantization (AF = 0.947), and on the attention-only side, where memory is spared, the memory-driven CFR gain does not appear at all (CFR *falls*

Table 5 Consolidated headline results: one row per finding with anchor set size n and key cells. The rising CFR is removed parametric competition (∼96%), not better context use (MLP-3b **0.43** vs. attn-3b **0.94**)

Finding	n	Key result (one-line claim)
Phenomenon (Qwen-3B)	96	Memory CB-Acc **1.00** → **0.84** while context CFR **0.60** → **0.74** (FP16→ 4-bit).
Phenomenon (Qwen-7B)	100	Down to 2-bit, CB-Acc **1.00** → **0.88**, CFR **0.74** → **0.85**; cleanest evidence.
Component mechanism (Qwen, RTN)	96	Quantizing the MLP erodes recall more than attention: **dCB** MLP **0.84** vs attn **0.14** (weight-mass-confounded).
Component mechanism (cross-quantizer)	96	MLP>attn survives a calibrated quantizer: GPTQ **dCB** MLP **0.62** vs attn **0.08**.
Comprehension (Qwen-3B)	160	Leak-free 2-hop survives 8-bit (**0.97**) and 4-bit (**0.74**); breaks at uniform 3-bit (**0.15**).
Method (Qwen-3B)	96	Mixed-3.6 matches uniform-4 functionality with higher CFR (**0.85** vs **0.74**) at fewer bits.
Recovery (Qwen-3B/7B)	96/100	Open-book = **1.00** to 2-bit (ceiling): single-hop copy-from-context is unimpaired.

from 0.604 to 0.521, a drop of −0.083; paired bootstrap 95% CI [−0.167, +0.000], n=96). The headline "context up" columns of Table 7 are therefore an honest description of *behavior* on conflict items, but they are not evidence that the model uses context better; the leak-free comprehension probe below is the result we lead with for context ability. We also checked that this attribution is not a decoding artefact: re-running the breakage-adjacent MLP-3b cell across greedy decoding plus three sampled seeds (temperature 0.7), CB-Acc is seed-stable (range [0.146, 0.156], sd 0.005) and CFR varies only modestly (sd 0.027). The headline recall collapse is seed-robust; the small CFR sampling spread is consistent with this cell being breakage-adjacent.

The most decisive demonstration comes from Qwen2.5-7B-Instruct with `llama.cpp` GGUF k-quants [51], where we can push to 2-bit while keeping garbage low (anchor n=100). As Table 7 (second block) and Fig. 2(b) show, from Q6 to Q2 memory drops from 1.00 to 0.88 while context-following climbs from 0.74 to 0.85, with OTHER < 0.08 at every step. The two curves separate cleanly and in opposite directions over a four-bit range, with no accompanying surge in garbage, exactly the signature of a selective erosion of stored knowledge rather than a generic loss of capability.

On CounterFact (anchor n=108), a standard external benchmark, the *direction* replicates: from FP16 to NF4, CB-Acc falls 1.00→0.82 while CFR rises 0.56→0.64 (Table 7, third block). However, its cloze format yields a high garbage rate (∼0.25), so its CFR is denominator-diluted; we therefore read CounterFact as supportive-but-noisy external evidence, not as a clean headline.

On PopQA [35], a standard entity-relation factual benchmark, the phenomenon replicates beyond our in-house probes. We cast 400 PopQA facts spanning sixteen relations into the same conflict format (type-correct, within-relation counterfactual swaps) and anchor on the n=164 items Qwen2.5-3B answers correctly closed-book. Table 6 shows the familiar split: as precision falls, closed-book recall erodes (CB-Acc 1.00 at FP16, 0.67 and 0.71 at 8- and 4-bit, 0.20 at uniform 3-bit) while copy-from-context stays at ceiling (open-book 0.99 through 4-bit). The component localization also replicates: MLP-only 3-bit quantization erases recall (CB-Acc 0.24, ΔCB = 0.76) far more than attention-only (CB-Acc 0.77, ΔCB = 0.23), with attention near-clean (OTHER = 0.06). We note two honest caveats: uniform 3-bit is a breakage regime on PopQA (OTHER = 0.45), and PopQA's long-tail facts are more fragile than FACTS_V2 (recall already drops to 0.67 by 8-bit). The source-fragmentation and the MLP locus nonetheless hold on a standard benchmark. Taken with CounterFact above, this means the headline is corroborated on

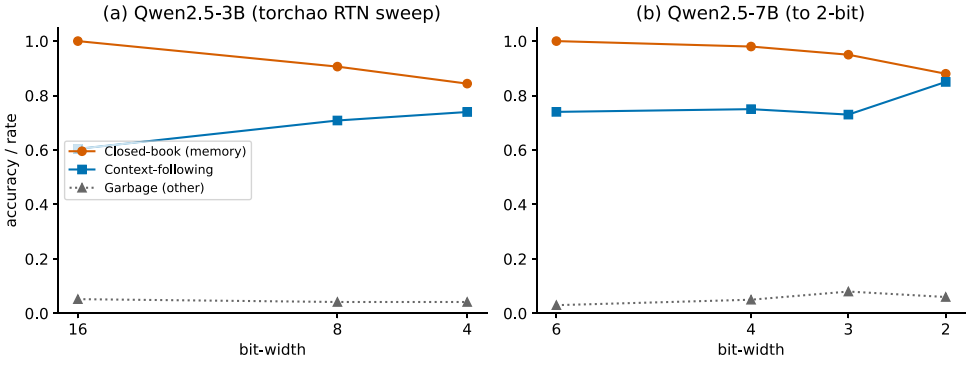


Fig. 2 CB-Acc, CFR, and garbage rate versus bit-width for (**a**) Qwen2.5-3B-Instruct and (**b**) Qwen2.5-7B-Instruct (GGUF ***k***-quants): CB-Acc falls, CFR is preserved or rises, and garbage stays low

Table 6 PopQA [35] standard benchmark (Qwen2.5-3B, anchor $n=164$): recall (CB-Acc) falls with bit-width while copy-from-context (open-book) holds; MLP-3 erases recall, attention-3 spares it. 95% Wilson CIs; uniform 3-bit is a breakage regime (OTHER > 0.2)

Config	Bits	CB-Acc [95% CI]	CFR [95% CI]	OTHER [95% CI]	open-book
FP16	16	1.00 [0.98,1.00]	0.82 [0.75,0.87]	0.02 [0.01,0.06]	0.99
8-bit	8	0.67 [0.60,0.74]	0.75 [0.68,0.81]	0.03 [0.01,0.07]	0.99
4-bit	4	0.71 [0.63,0.77]	0.91 [0.85,0.94]	0.02 [0.01,0.05]	0.99
3-bit (uniform)	3	0.20 [0.14,0.26]	0.54 [0.47,0.62]	0.45 [0.37,0.52]	0.62
MLP-only 3-bit	3	0.24 [0.18,0.31]	0.71 [0.63,0.77]	0.24 [0.18,0.31]	0.82
Attn-only 3-bit	3	0.77 [0.70,0.83]	0.66 [0.58,0.73]	0.06 [0.03,0.11]	0.99

Table 7 Quantization selectively erases memory: in the functional regime (OTHER < 0.2), CB-Acc falls with bit-width while CFR holds. Brackets give 95% Wilson CIs. CounterFact replicates (high garbage); Llama-3.2-3B is a no-erosion boundary

Config	Bits	CB-Acc [95% CI]	CFR [95% CI]	OTHER [95% CI]
Qwen2.5-3B-Instruct (facts_v2, anchor $n=96$)				
FP16	16	1.00 [0.96,1.00]	0.60 [0.50,0.70]	0.05 [0.02,0.12]
INT8 (bnb)	8	0.97 [0.91,0.99]	0.64 [0.54,0.72]	0.05 [0.02,0.12]
8-bit (torchao)	8	0.91 [0.83,0.95]	0.71 [0.61,0.79]	0.04 [0.02,0.10]
NF4	4	0.95 [0.88,0.98]	0.71 [0.61,0.79]	0.05 [0.02,0.12]
4-bit (torchao)	4	0.84 [0.76,0.90]	0.74 [0.64,0.82]	0.04 [0.02,0.10]
Qwen2.5-7B-Instruct (GGUF k-quants, anchor $n=100$)				
Q6	∼6	1.00 [0.96,1.00]	0.74 [0.65,0.82]	0.03 [0.01,0.08]
Q4	4	0.98 [0.93,0.99]	0.75 [0.66,0.82]	0.05 [0.02,0.11]
Q3	3	0.95 [0.89,0.98]	0.73 [0.64,0.81]	0.08 [0.04,0.15]
Q2	2	0.88 [0.80,0.93]	0.85 [0.77,0.91]	0.06 [0.03,0.12]
CounterFact (external, anchor $n=108$; high garbage)				
FP16	16	1.00 [0.97,1.00]	0.56 [0.46,0.65]	0.25 [0.18,0.34]
NF4	4	0.82 [0.74,0.88]	0.64 [0.54,0.72]	0.22 [0.15,0.31]
Llama-3.2-3B-Instruct (boundary: no erosion)				
FP16	16	1.00	0.77	—
NF4	4	1.00	0.75	—

two standard external factual benchmarks (PopQA and CounterFact) in addition to the two capability controls drawn from standard suites (MMLU and WikiText-2), so the central result does not rest on our in-house probes alone; the in-house probes add what public benchmarks cannot, a guaranteed leak-free measurement of context use.

The phenomenon is model- and quantizer-dependent. On Llama-3.2-3B-Instruct [49], BNB 4-bit *preserves* closed-book knowledge: CB-Acc is 1.00 at both FP16 and NF4, and CFR is essentially flat ($0.77 \rightarrow 0.75$). When memory does not erode there is no knowledge–context trade to observe, and the phenomenon does not arise. We report Llama honestly as a negative/boundary result rather than as supporting evidence (Table 7, last block), and return to its implications in Sect. 5.

This boundary case deserves emphasis rather than a footnote, because it directly shapes how the paper's findings should be applied. The memory-versus-context trade that PRISM measures can only manifest *when memory actually erodes*; a model–quantizer pairing that preserves closed-book recall, here Llama-3.2-3B under 4-bit `bitsandbytes` NF4, exhibits no phenomenon by construction, not because the framework fails but because there is no degradation to fragment. The practical implication is that our quantitative floors are stated for the pairings we measured and should not be assumed to transfer unchanged: a practitioner who reads "MLPs are fragile, protect their

precision" and then quantizes a Llama checkpoint may observe no recall loss at all and conclude the effect is illusory. We therefore state plainly that the safe course is to run PRISM's two probes on the specific model and quantizer being deployed, rather than to port the Qwen numbers across architectures. As to *why* Llama is a boundary, we can only offer candidates for future work to disentangle: NF4's learned quantization grid may happen to align with Llama's weight distribution, or architectural and pretraining differences may leave Llama's early-MLP recall less precision-critical than Qwen's. Adjudicating among these is beyond the single causal instrument we deploy here, and we flag it as an open question rather than resolving it.

All headline rates are computed on the anchor set, which pins the full-precision baseline at CB-Acc = 1.00 by construction (Sect. 3). To rule out the worry that anchoring manufactures the trend, we also run *unfiltered* (full-set) sweeps and find the same recall-versus-context pattern with nearly identical magnitudes: for example, the component MLP-3b CFR reads 0.823 anchored versus 0.807 unfiltered. The effect is a property of the quantization, not of the anchoring; we report the unfiltered figures as a robustness check in Sect. 5.

Because the CFR "context up" columns reflect removed competition rather than better context use (above), our *primary* evidence on context ability is the leak-free comprehension probe of this subsection, not CFR. It directly measures whether the model can use supplied context with no parametric competitor to confound the reading: closed-book accuracy is pinned at 0.00 for every configuration, so any score is purely contextual. A reasonable objection to the open-book results is that following a supplied passage may be mere verbatim *copying* rather than genuine *comprehension*, and that the surviving capability is therefore trivial. We address this with a probe designed to be leak-free and to demand composition rather than copying. The task is a synthetic two-hop question over *fictional* entities (so no fact can be recalled from pretraining), with distractor sentences in the context, requiring the model to chain two supplied facts to produce an answer that appears *nowhere verbatim*. The critical control is the closed-book condition: with no context, accuracy is *exactly*0.00 for *every* configuration (Table 8), confirming there is no leakage, any accuracy is purely contextual comprehension, not memory and not copying. Figure 3 visualizes these results: accuracy against bit-width in panel (a) and the component-wise 3-bit split in panel (b).

Three findings follow. First, genuine comprehension is preserved by moderate quantization: it survives 8-bit nearly intact (0.98 → 0.97) and largely survives 4-bit (0.74), so the open-book robustness reported above is not an artefact of trivial copying, the model is composing supplied facts, not echoing them. Second, multi-hop comprehension is *more fragile* than single-hop verbatim copy: where copy-from-context held at ceiling down

Table 8 Leak-free two-hop comprehension (Qwen2.5-3B-Instruct, fictional entities, n=160; closed-book **0.00**, 95% Wilson CIs): survives 8-bit and largely 4-bit, breaks at uniform 3-bit; MLP breaks it, attention preserves it

Config	Closed-book	Comprehension (support) [95% CI]	Conflict-follow
FP16	0.00	0.98 [0.95,0.99]	0.96
8-bit	0.00	0.97 [0.93,0.99]	0.94
4-bit	0.00	0.74 [0.66,0.80]	0.78
3-bit (uniform)	0.00	0.15 [0.10,0.21]	0.13
MLP-only 3-bit	0.00	0.43 [0.36,0.51]	0.44
Attn-only 3-bit	0.00	0.94 [0.89,0.97]	0.91

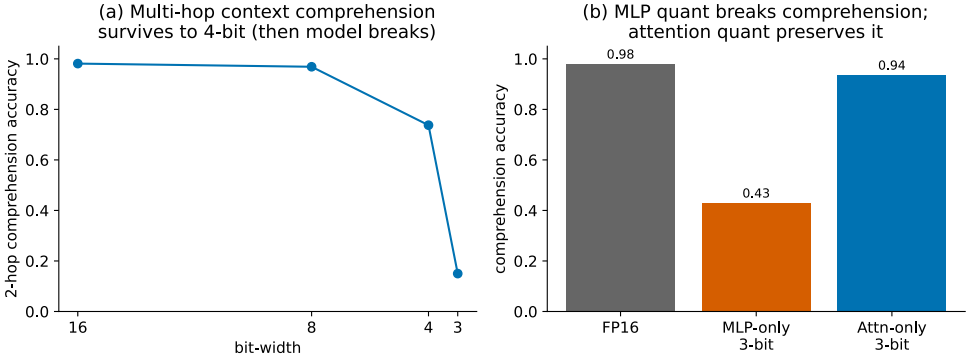


Fig. 3 Leak-free multi-hop comprehension (Qwen2.5-3B-Instruct, closed-book = **0.00**): (**a**) survives 8-bit and largely 4-bit, breaks at uniform 3-bit; (**b**) At 3-bit MLP breaks it (**0.98** → **0.43**), attention preserves (**0.94**)

to 2-bit (Sect. 4), composition breaks at uniform 3-bit (0.15), because two-hop chaining is exactly the kind of heavy MLP computation that collapses when the MLP is pushed below its operating floor. Third, and most tellingly, the component-wise split tracks the mechanism: MLP-only 3-bit quantization *breaks* comprehension (0.98 → 0.43) while attention-only 3-bit quantization *preserves* it (0.94). The MLP performs the compositional computation; attention is robust. This both refutes the "it's just copying" critique and refines our central theme: *MLP-dependent* capabilities, parametric recall *and* multi-hop reasoning, are precision-fragile, whereas *attention-dependent* simple context use is precision-robust.

Taken together, the Qwen results establish a robust, garbage-controlled trend: in the functional regime, quantization erodes parametric memory while leaving the ability to read supplied context comparatively intact. We are careful about the direction of the claim: the rising CFR on conflict items is the *disappearance of the parametric competitor*, not improved context use; the positive evidence that context ability survives is the leak-free comprehension probe, where accuracy is purely contextual (closed-book = 0.00) and genuine two-hop composition survives 8-bit and largely 4-bit. The practical consequence is that closed-book benchmarks, which probe exactly the parametric memory that quantization preferentially erases, *overstate* the deployment harm of quantization for context-grounded, retrieval-augmented use. A model that has lost closed-book facts to 2-bit compression may nonetheless read and compose supplied context about as well as its full-precision counterpart, at least down to 4-bit. The next section asks *why*, by causally localizing this trade-off to specific transformer sublayers.

In summary, in the functional regime, low-bit quantization erodes parametric memory (Qwen-3B CB-Acc 1.00 → 0.84; Qwen-7B 1.00 → 0.88 at 2-bit). The accompanying rise in context-following on conflict items is *removed competition*, not improved context use (96% of the MLP-3b CFR gain comes from items that also lost their memory; on the memory-intact subset CFR barely moves, 0.667 → 0.733); our primary context-ability evidence is instead the leak-free comprehension probe (closed-book = 0.00), where genuine two-hop composition survives to 4-bit and the component split tracks the mechanism (MLP-3b 0.43 vs attn-3b 0.94). Closed-book benchmarks therefore overstate the deployment harm of quantization for retrieval-augmented use.

Section 4 established the paper's headline as a *phenomenon*: as average precision falls, open-ended parametric recall (CB-Acc) is the most fragile capability, verbatim copy-from-context the most robust, and multi-hop comprehension intermediate, a pattern

that is quantizer- and family-general. This section asks a narrower, mechanistic question: *which component* of the network carries the precision-fragile recall? We answer it with a component-level causal probe, but we are deliberately careful about how far the answer generalizes. Using bit-width as a targeted instrument, holding architecture, prompts, decoding, and anchor set fixed and quantizing *only* the MLP/FFN projections, or *only* the attention projections, or depth-restricted bands of the MLP, with the complementary weights (and `lm_head`) left in FP16, we establish a robust component-level result and then stress-test it against two confounds that constrain its interpretation.

The robust, reproducible claim is component-level: quantizing the MLP erodes parametric recall far more than quantizing attention, and this survives a calibrated (GPTQ) quantizer and replicates across three model families (Qwen2.5-3B, Llama-3.2-3B, Gemma-2-2B). We then report two honest temperings that we do *not* hide. (i) The component effect is *weight-mass-confounded*: the MLP holds $\sim$88% of the non-embedding parameters, so equal bit-width is not equal perturbation. A weight-mass control *refutes* this confound under naive round-to-nearest (RTN) quantization, at equal perturbed mass the early MLP still erases far more recall than attention, but the very same equal-mass effect *vanishes* under calibrated GPTQ (early-MLP dCB = 0.000). We therefore do *not* claim a quantizer-general per-parameter localization; the mass-controlled early-MLP result is specific to RTN (and, in its depth, to Qwen). (ii) Even where recall is erased, MLP quantization *broadly* degrades the model (perplexity, MMLU), so the effect is graded, not surgical. The honest takeaway is a component-level statement, the MLP is the locus of the perturbation-induced recall collapse; attention is precision-robust, carried by a single causal instrument and qualified by the mass and calibration caveats below.

The primary intervention uses the Qwen2.5-3B-Instruct model and the `facts_v2` anchor set ($n = 96$) from the phenomenon study. We run two quantizers: the naive `torchao` round-to-nearest (RTN) integer quantizer of the phenomenon study, and, as a calibrated alternative, GPTQ (256 wikitext-2 calibration sequences) with `gptqmodel`'s `dynamic` skip-non-target so that only the intended modules are quantized (targeting verified: 108 MLP / 144 attention / 15 early-MLP / 15 mid-MLP modules). Component filters select the MLP-only projections (gate/up/down), the attention-only projections (q/k/v/o), or depth-restricted slices of the MLP (e.g. the first five or first six layers). The filters are verified disjoint, and `lm_head` is kept in FP16 in every configuration so that the output-embedding contribution cancels across conditions. We read the localization at 3-bit: pushing the MLP to 2-bit kills the whole model (CB-Acc, CFR, and open-book accuracy all collapse to 0), because the MLP also performs essential computation, so 3-bit is the operating floor for the intervention.

A component result obtained with a naive RTN quantizer invites two related worries: that it reflects RTN's error profile rather than a property of the network, and that equal bit-width is not equal perturbation, since the MLP carries $\sim$88% of the non-embedding parameters and a larger MLP effect could simply reflect more perturbed weight-mass. We address both confounds here. First we repeat the component-wise intervention with a strong calibrated quantizer (GPTQ) and compare it head-to-head with RTN at the same 3-bit budget; then we hold the *perturbed mass* fixed and vary *which* layers absorb it. The comparison is decisive in *two* directions, and we report both honestly. Table 9 and Fig. 4 give the numbers.

Table 9 Cross-quantizer component quantization (Qwen2.5-3B, 3-bit,$n = 96$): MLP> attention survives GPTQ (all-MLP**0.615**vs. attention**0.083**); early-MLP localization is RTN-specific (GPTQ$\mathbf{dCB} = 0.000$)

Perturbed (3-bit)	RTNdCB	GPTQdCB
all-MLP	**0.844**	**0.615**
all-attention	**0.135**	**0.083**
EARLY-5 MLP (∼**340**M)	**0.667**	**0.000**
MID-5 MLP (∼**340**M)	**0.010**	**0.042**

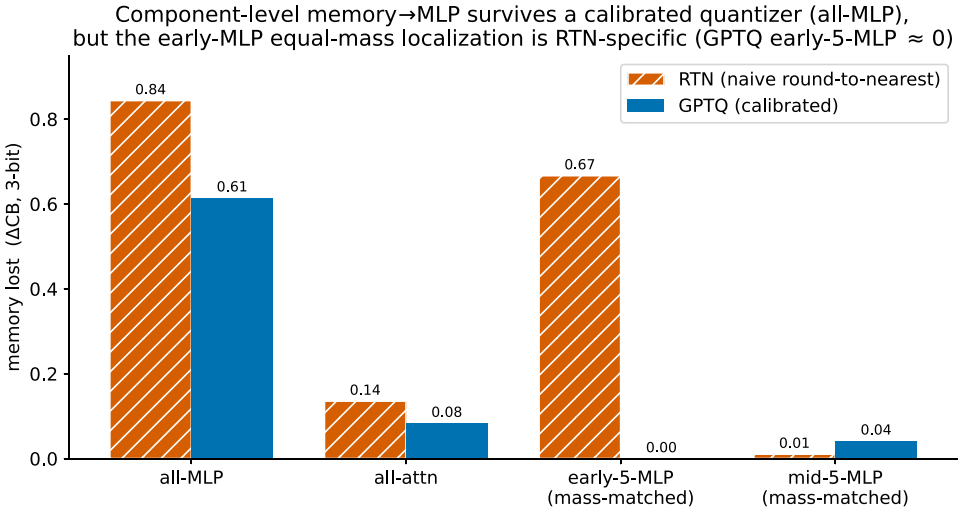


Fig. 4 Closed-book loss **dCB**, RTN vs. GPTQ (Qwen2.5-3B, 3-bit, $n = 96$): all-MLP (**0.844**/**0.615**) > attention (**0.135**/**0.083**); early-5-MLP **0.667** vanishes under GPTQ (**0.000**)

The component MLP>attention effect survives calibration. Quantizing all MLP projections erodes recall far more than quantizing all attention under both quantizers: RTN dCB = 0.844 vs. attention 0.135, and GPTQ dCB = 0.615 (CB-Acc 0.385, support 0.93, OTHER = 0.08) vs. attention 0.083 (CB-Acc 0.917). GPTQ's calibration shrinks the absolute erasure, as expected, a smarter quantizer preserves more, but it does not change the *component ordering*: the MLP is the locus of the recall collapse under both the naive and the calibrated quantizer. This is the robust, quantizer-general component-level claim.

It is worth being explicit about *why* a calibrated quantizer degrades recall less, and less unevenly, than naive round-to-nearest, since this is what separates the quantizer-general component result from the round-to-nearest-specific early-layer refinement. Round-to-nearest maps each weight to the closest grid point at a fixed per-group scale, so its error is largest exactly where the weight distribution is heavy-tailed or high-variance, which the early MLP layers of Qwen tend to be, and that error is applied blindly regardless of a weight's influence on the output. GPTQ instead fits per-group scales and zero-points to minimize a calibrated reconstruction objective on WikiText-2 activations, spending its limited bit budget preferentially on the weights whose perturbation most changes the layer output. The consequence is twofold and matches the numbers: the *absolute* erasure falls under calibration (all-MLP ΔCB drops from 0.844 to 0.615), and the *within-MLP* concentration that round-to-nearest produces in the early layers is smoothed away entirely (equal-mass early-5 MLP ΔCB from 0.667 to 0.000), because calibration protects precisely those high-salience early-MLP weights that round-to-nearest damages

Table 10 Weight-mass control (Qwen2.5-3B, RTN 3-bit, $n = 96$): at equal ∼340M mass, early-MLP erases far more recall than attention or middle-MLP; RTN-specific (cf. cross-quantizer Table 9)

Perturbed (∼340 M)	Bits	CB-Acc [95% CI]	dCB	Support(copy)	OTHER
(none) FP16	16	**1.00** [0.96,1.00]		**1.00**	**0.05**
EARLY-5 MLP	3	**0.333** [0.25,0.43]	**0.667**	**0.896** [0.82,0.94]	**0.052** [0.02,0.12]
ALL-ATTENTION	3	**0.865** [0.78,0.92]	**0.135**	**1.000**	**0.05**
MID-5 MLP	3	**0.990** [0.94,1.00]	**0.010**	**1.000**	**0.02**

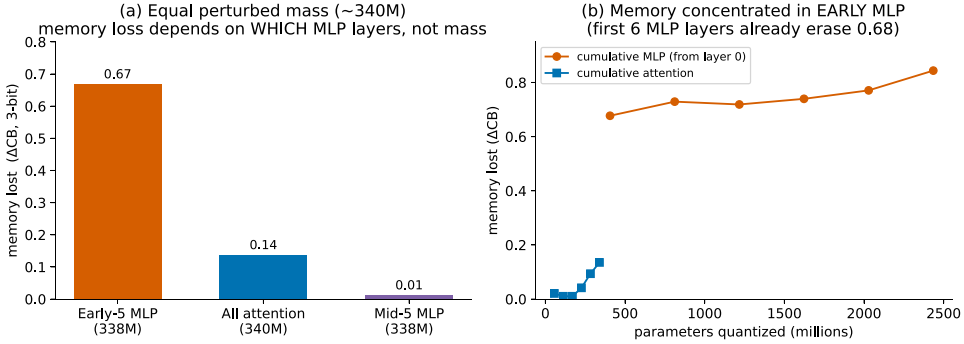


Fig. 5 Under RTN (Qwen2.5-3B, 3-bit, $n = 96$), at ∼340M equal mass early-MLP dCB = 0.667 vs. attention 0.135, middle-MLP 0.010; first six MLP layers give 0.677 of the 0.844 all-MLP total; RTN-specific

indiscriminately. The *component-level* ordering nonetheless survives calibration because the MLP still absorbs the bulk of the reconstruction error under any weight-only scheme, being where most of the parameters and most of the factual computation sit. This is the mechanistic reason we present the component-level result as quantizer-general while scoping the early-layer localization to round-to-nearest.

The mass-controlled early-MLP localization, however, is RTN-specific. The equal-mass comparison tells a different, more cautionary story across quantizers. Restricting the perturbation to the first five MLP layers (∼340M parameters) erases dCB = 0.667 under RTN but dCB = 0.000 under GPTQ (CB-Acc 1.000); the mass-matched middle-MLP band is negligible under both (0.010 RTN, 0.042 GPTQ). In other words, the equal-mass *refutation* of the weight-mass confound that we present next is an RTN-only result: under calibrated GPTQ there is *no* mass-controlled MLP>attention localization, at equal perturbed mass, GPTQ's early-MLP quantization leaves recall intact while attention (whose ∼340M all-projection mass GPTQ erodes by 0.083) actually erases more. We therefore do not claim a quantizer-general per-parameter localization. The early-MLP, mass-controlled result is specific to naive RTN and, in its depth, to Qwen; the quantizer-general claim is strictly component-level (whole-MLP vs. whole-attention).

We now hold the perturbed mass fixed and vary which layers absorb it, scoping every claim in this control explicitly to the naive RTN quantizer: as the cross-quantizer comparison makes clear, this control refutes the confound *under RTN* but not under calibrated GPTQ, so we treat it as evidence that the RTN component effect is not *merely* a mass artefact, not as a quantizer-general per-parameter localization. At 3-bit (RTN) we match the number of perturbed parameters across three configurations at ∼340M weights: the first five MLP layers (EARLY-5 MLP), all attention projections (ALL-ATTENTION), and a contiguous block of five *middle* MLP layers (MID-5 MLP). Table 10 and Fig. 5 report the result. At equal perturbed mass, EARLY-5 MLP erases dCB = 0.667 of closed-book memory (CB-Acc 1.00 → 0.333), whereas ALL-ATTENTION erases only 0.135

(CB-Acc 0.865) and MID-5 MLP erases a negligible 0.010 (CB-Acc 0.990). Under RTN, early-MLP quantization thus erases roughly 5× more memory than the same mass spent on attention and ∼70× more than the same mass spent on middle MLP layers. Because the perturbed mass is held constant, the RTN effect cannot be a pure weight-mass artefact: *where* the mass sits, specifically in the early MLP, is what drives the RTN recall collapse. We stress that this equal-mass refutation does *not* hold under GPTQ (Table 9, early-5-MLP dCB = 0.000), so it is RTN-scoped.

Depth and component cannot be fully disentangled by mass (M2). Mass cannot be matched symmetrically: the early-5 attention projections total only ∼47M parameters, far below the ∼340M early-5-MLP mass, so an equal-mass "early attention" comparator does not exist architecturally. The honest within-depth control is therefore *layer-matched* rather than mass-matched: restricting both components to the same first six layers, the MLP erases dCB = 0.677 versus attention's 0.021. This layer-matched comparison, like the equal-mass one, holds under RTN; under GPTQ even the layer-restricted MLP effect is mass-consistent (no mass-controlled MLP>attention). We accordingly soften the interpretation: rather than claiming "the MLP *stores* the fact," we say that under this perturbation the MLP is the *locus of the perturbation-induced recall collapse*.

The RTN early-MLP erasure is also selective. Under RTN, the early-MLP intervention that destroys memory leaves the model's other capabilities largely intact: on supported items its copy-from-context (support) accuracy is 0.896 and its garbage rate is only OTHER = 0.052 (Table 10), so the model is not merely broken, it has lost the *fact* while still reading and copying from context coherently. This is the selective signature we cannot obtain by perturbing equal mass anywhere else in the network under RTN.

The early-layer concentration under RTN is corroborated by a cumulative sweep (Fig. 5): quantizing the first six MLP layers already erases dCB = 0.677 of the 0.844 erased by quantizing *all* MLP layers, whereas the cumulative attention curve saturates at ≤ 0.135 . Under RTN, the bulk of the parametric recall that quantization can destroy is concentrated in the bottom of the MLP stack. Two further RTN controls, requiring no new mass-matching, point the same way. Control C (harder bits on attention). Even when we make the attention perturbation *harder* than the MLP perturbation, quantizing attention to 2-bit versus the MLP to 3-bit, attention retains *more* memory (CB-Acc 0.354, 95% CI $[0.27, 0.45]$) than the MLP (CB-Acc 0.156, 95% CI $[0.10, 0.24]$). A more aggressive bit budget on attention still erases less memory than a milder budget on the MLP. Control C' (equal measured dequantization error). Matching the *measured* relative dequantization error rather than the bit-width or the mass, both components at relative error 0.177 (the 3-bit operating point), the MLP still erases dCB = 0.844 versus attention's 0.135: the two components do not collapse onto a single error–memory curve, so the RTN gap is not explained by how much numerical error each weight set absorbs. Together, the equal-mass comparison, the cumulative sweep, and Controls C/C' show that under RTN the recall collapse is not merely a weight-mass artefact: under this naive quantizer, the MLP, and specifically its early layers, is the locus of the perturbation-induced recall collapse.

We are precise, finally, about what is and is not claimed at the per-parameter level. The mass-controlled localization is specifically about the *early* MLP, specifically under *RTN*, and specifically on Qwen. Even within RTN, per-parameter sensitivity is *not* uniformly higher in the MLP than in attention: the mass-matched MID-5 MLP barely moves memory (dCB = 0.010), and there is no single dominant layer (every individual

Table 11 Memory-to-MLP localization replicates across families (`torchao` 3-bit): MLP erodes recall more than attention in Qwen2.5-3B, Llama-3.2-3B, Gemma-2-2B; early-layer specificity is Qwen-only (Gemma early-5 $\mathbf{dCB = 0.019}$)

Family (n)	Quantize	Bits	CB-Acc [95% CI]	dCB	Support(copy)
Qwen2.5-3B (**96**)	(none) FP16	16	**1.00** [0.96,1.00]	-	**1.00**
	MLP-only	3	**0.16** [0.10,0.24]	**0.84**	**1.00**
	Attn-only	3	**0.86** [0.78,0.92]	**0.14**	**1.00**
Llama-3.2-3B (**92**)	(none) FP16	16	**1.00** [0.96,1.00]	-	**1.00**
	MLP-only	3	**0.62** [0.52,0.71]	**0.38**	**1.00**
	Attn-only	3	**0.97** [0.91,0.99]	**0.03**	**1.00**
Gemma-2-2B (**105**)	(none) FP16	16	**1.00** [0.96,1.00]	-	**1.00**
	MLP-only	3	**0.638** [0.54,0.72]	**0.362**	**1.000**
	Attn-only	3	**0.981** [0.93,0.99]	**0.019**	**1.000**

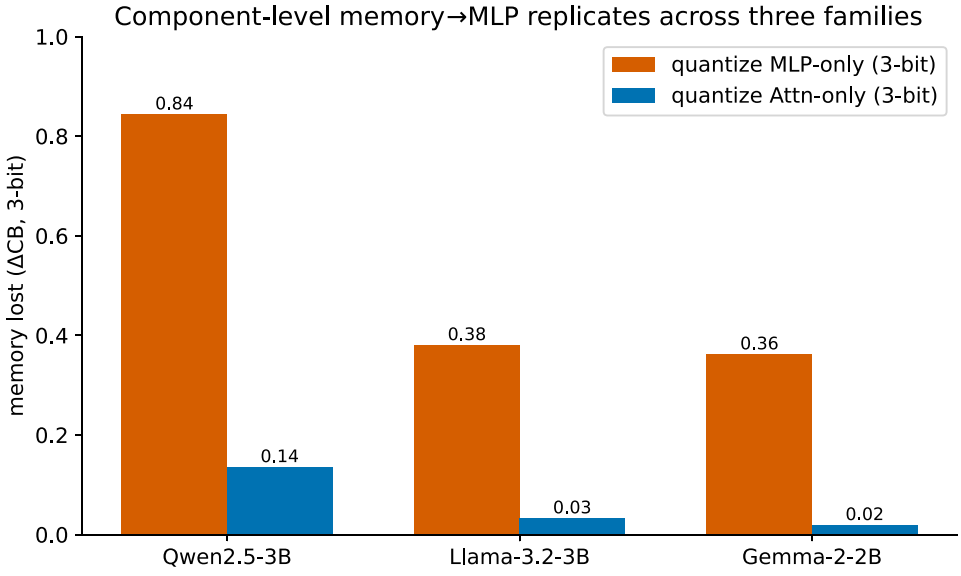


Fig. 6 Cross-family memory loss dCB (`torchao` 3-bit): MLP>attention for Qwen2.5-3B (**0.84** vs. **0.14**), Llama-3.2-3B (**0.38** vs. **0.03**), Gemma-2-2B (**0.36** vs. **0.02**); early-layer depth is Qwen-specific

layer contributes ≤ 0.021). Normalized per-100 M parameters, the RTN MLP and attention sensitivities are in fact comparable ($\sim$0.015 vs $\sim$0.022); the large MLP effect at the whole-component level comes from the *early layers*, not from a blanket per-parameter superiority of MLP weights. And as the cross-quantizer comparison above shows, the equal-mass early-MLP effect itself disappears under calibrated GPTQ. We therefore make *no* claim of a quantizer-general per-parameter MLP>attention localization. The quantizer-general, cross-family claim is strictly *component-level* (whole-MLP vs. whole-attention); the early-layer, mass-controlled refinement is a naive-RTN, Qwen-specific observation, and even the depth of that hotspot is model-specific (it is the early MLP on Qwen but not on Gemma, as the cross-family results below show).

A single-model causal claim invites the worry that it reflects an idiosyncrasy of one network. We therefore repeat the identical component-wise quantization on two further model families, Llama-3.2-3B-Instruct (anchor $n = 92$) and Gemma-2-2B-it (anchor $n = 105$), each quantized only on its MLP or only on its attention. Table 11 and Fig. 6 show that the *component-level* memory localization replicates in all three families: at

3-bit, MLP quantization erodes closed-book memory far more than attention quantization in every case.

On Qwen, MLP quantization collapses CB-Acc from 1.00 to 0.16 (dCB = 0.84) while attention spares it (1.00 → 0.86, dCB = 0.14); Figure 8 shows the full Qwen component-wise picture, with copy-from-context preserved under MLP quantization at both 3-bit and 4-bit. On Llama, MLP quantization erodes CB-Acc from 1.00 to 0.62 (dCB = 0.38, 95% CI [0.29, 0.48]) while attention spares it (1.00 → 0.97, dCB = 0.03, 95% CI [0.01, 0.09]). On Gemma, MLP quantization erodes CB-Acc to 0.638 (dCB = 0.362, 95% CI on CB-Acc [0.54, 0.72]) while attention is nearly lossless (0.981, dCB = 0.019), with copy-from-context support = 1.000 and an even cleaner garbage rate (OTHER ≈ 0.01) than Qwen. Because the FP16 anchor is pinned at CB-Acc 1.00, each dCB interval follows directly from the Wilson CI on the post-quantization CB-Acc, and in every family the MLP and attention intervals are non-overlapping. We treat dCB itself as the effect size of a quantization intervention (the absolute drop in retained recall from full precision), and read the non-overlapping 95% intervals as the significance statement for the MLP-versus-attention contrast: in all three families the MLP effect exceeds the attention effect by a margin far larger than either interval's width (Qwen 0.84 vs. 0.14, Llama 0.38 vs. 0.03, Gemma 0.36 vs. 0.02), so the ordering is not attributable to sampling variation. Where the comparison is paired on a shared anchor set we additionally report paired percentile-bootstrap intervals ($B = 10000$) rather than the marginal Wilson intervals. Across three families, the precision-fragility of parametric memory is concentrated in the MLP and attention is precision-robust, the central, reproducible mechanistic claim of this paper.

While the component-level localization is cross-family robust, the *depth* of the hotspot is model-specific. On Gemma-2-2B, restricting the perturbation to the first five MLP layers erases only dCB = 0.019, the early MLP is *not* Gemma's memory hotspot, even though Gemma's whole-MLP quantization erases dCB = 0.362. The early-layer specificity is therefore a Qwen property, not a universal one. We report the cross-family claim precisely as a *component-level* statement (the MLP is the locus of the recall collapse, attention is robust) and flag the early-layer depth as model-specific.

A whole-set localization could in principle be driven by a single dominant relation rather than holding broadly. We therefore disaggregate the Qwen RTN component-wise intervention by the five fact relations in `facts_v2`. Table 12 shows that the component localization holds in *every* relation: MLP quantization erodes closed-book recall by dCB = 0.71–1.00, the early-MLP variant by 0.52–1.00, while attention quantization stays low throughout (dCB = 0.06–0.23). The *direction* is uniform (MLP ≫ attention in all five relations); the *magnitude* varies across relations. Per-relation sample sizes are small ($n = 12$–31), so the Wilson intervals are wide and we present the table as a uniform-direction robustness check rather than a precise per-relation effect-size estimate.

Table 12 Per-relation closed-book loss **dCB** (Qwen2.5-3B, RTN 3-bit) for MLP-only, early-MLP, attention-only across five relations: MLP dominates attention uniformly; small ***n***, wide 95% Wilson intervals

Relation	n	ΔCB MLP-3b	ΔCB early-MLP	ΔCB attn-3b
Company CEO	13	1.00 [0.77,1.00]	1.00 [0.77,1.00]	0.23 [0.08,0.50]
Company HQ city	24	0.71 [0.51,0.85]	0.62 [0.43,0.79]	0.12 [0.04,0.31]
Composer nationality	12	0.75 [0.47,0.91]	0.58 [0.32,0.81]	0.08 [0.01,0.35]
Film director	16	1.00 [0.81,1.00]	0.81 [0.57,0.93]	0.06 [0.01,0.28]
Founding year	31	0.84 [0.67,0.93]	0.52 [0.35,0.68]	0.16 [0.07,0.33]

Table 13 Graded, not surgical (Qwen2.5-3B, 3-bit): MLP quantization raises PPL **4.46×** and halves MMLU; conflict recall most fragile, support most robust. FP16 reference: recall **0.842**, MMLU **0.657**, PPL **10.88**

Config	Recall [95% CI]	Δrecall	MMLU [95% CI]	ΔMMLU	PPL ratio
FP16	**0.842** [0.76,0.90]		**0.657** [0.64,0.68]		**1.00×**
MLP-3b	**0.131** [0.08,0.21]	**0.711**	**0.306** [0.29,0.33]	**0.351**	**4.46×**
early-MLP-3b	**0.298** [0.22,0.39]	**0.544**	**0.384** [0.36,0.41]	**0.273**	**3.27×**
Attn-3b	**0.772** [0.69,0.84]	**0.070**	**0.633** [0.61,0.65]	**0.024**	**1.08×**

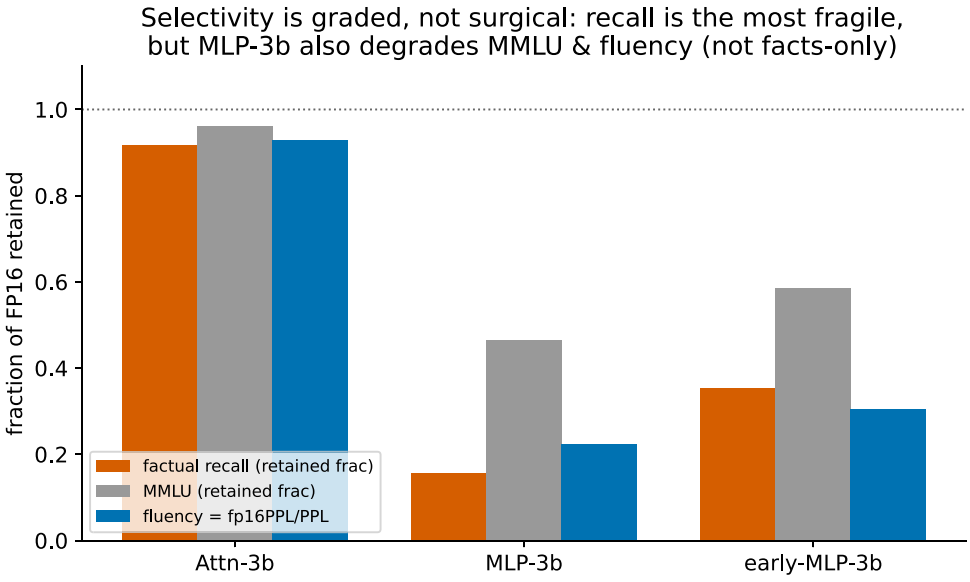


Fig. 7 Graded selectivity (Qwen2.5-3B, 3-bit): MLP-3b erases recall most ($\Delta = 0.711$), raises PPL **4.46×**, halves MMLU; support survives at **1.00**; attention-3b is the near-flat control

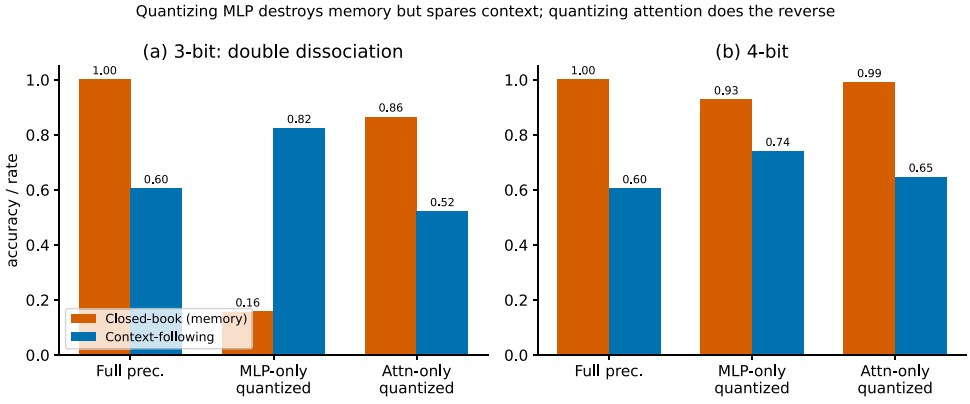


Fig. 8 Component-wise quantization (Qwen2.5-3B), CB-Acc vs. copy-from-context at (**a**) 3-bit and (**b**) 4-bit: MLP-only erases memory, attention-only spares it; survives GPTQ (Fig. 4) and replicates across families (Fig. 6)

Localizing memory loss to the MLP raises an honest question: when MLP quantization erases recall, does it erase *only* facts, or does it broadly damage the model? We answer with a selectivity control that measures, alongside conflict recall, two capabilities that are orthogonal to parametric facts, general knowledge (MMLU, $n = 1995$) and language-modeling fluency (wikitext-2 perplexity). Table 13 and Fig. 7 report the result, and the answer is honestly graded: MLP quantization is *not* a surgical fact-eraser.

At 3-bit, MLP quantization drops conflict recall by $\Delta = 0.711$ (from 0.842 to 0.131), but it also raises perplexity by 4.46× and roughly halves MMLU ($\Delta = 0.351$, from 0.657 to 0.306). So MLP quantization broadly degrades the model; it does not leave everything but facts intact. What survives the ordering, however, is the key finding: *recall is the most fragile capability and copy-from-context the most robust*. The same MLP-3b model that loses 71% of its recall retains support (copy) accuracy of 1.00; MCQ knowledge and fluency degrade by intermediate amounts. The early-MLP-only variant sits between, with smaller drops on all three axes (Δrecall = 0.544, ΔMMLU = 0.273, PPL 3.27×). Attention quantization at 3-bit is the near-flat negative control: it barely moves any axis (Δrecall = 0.070, ΔMMLU = 0.024, PPL 1.08×). The takeaway is therefore an *ordering* of fragility, recall > MCQ/fluency > copy, not a claim that facts are excised while the rest of the model is untouched.

This component-level result connects to a known mechanistic picture along a new axis. Prior work shows that factual associations are stored in and editable through the MLP/FFN sublayers, the feed-forward layers act as key–value memories [12], individual facts can be located and rewritten there [13], and specific "knowledge neurons" in the FFN gate factual recall [18]. Those results manipulate *values* (editing weights to change which fact is recalled). We add an orthogonal manipulation of *precision*: reducing the numerical resolution of the MLP sublayers degrades parametric recall while attention remains precision-robust, across a calibrated quantizer and three families. We are careful not to overclaim: this shows the MLP *matters* for recall under perturbation, consistent with the literature on where facts live, but it is weaker than the claim that the MLP *stores* the fact. The cross-quantizer, cross-family component localization turns a correlational compression trend into a reproducible component-level causal statement, and motivates the precision-allocation method of Sect. 4.

Several qualifications keep the claim precise. First, the localization is read at 3-bit because the MLP is not *only* memory: pushing the MLP to 2-bit kills the whole model (CB-Acc, CFR, and open-book accuracy collapse to 0), since the MLP also performs essential computation; the MLP is the precision-fragile seat of *both* parametric memory *and* heavy computation, consistent with the comprehension probe of Sect. 4 where MLP quantization breaks multi-hop reasoning. Second, the mass-controlled per-parameter localization is *not* quantizer-general: the equal-mass early-MLP effect holds under RTN but vanishes under calibrated GPTQ (Table 9), and even within RTN the mass-matched middle MLP barely matters while per-100 M MLP and attention sensitivities are comparable. The quantizer-general claim is therefore strictly component-level (whole-MLP vs. whole-attention); the early-MLP, mass-controlled refinement is a naive-RTN, Qwen-specific observation, and the depth is itself model-specific (Qwen, not Gemma). Third, selectivity is graded, not surgical: at the bits where recall fails, perplexity and MMLU also degrade, recall is simply the *most* fragile capability and copy-from-context the most robust. Fourth, the intervention is run at the 3B/2B scale because component-wise quantization requires an FP16 base on a single 8 GB GPU. Fifth, the causal evidence rests on a *single* instrument, component-wise quantization. "Quantizing the MLP destroys recall" establishes that the MLP *matters* for recall under perturbation; it is weaker than "the MLP *stores* the fact." We report a complementary activation-patching test, which is confounded by an FP16↔quantized distribution mismatch, only as a methodological note (see Appendix); we rely on the weight-level component-wise quantization as the cleaner

causal instrument, since reducing the precision of the targeted weights is a direct causal manipulation of exactly the component whose contribution we measure, sidestepping the known gap between where a fact can be *localized* and where an *edit* most changes behaviour [47]. The component evidence should accordingly be read as single-instrument causal support, not as a multi-method triangulation.

In summary, the robust, reproducible mechanistic statement is *component-level*: quantizing the MLP erodes parametric recall far more than quantizing attention, and this survives a calibrated GPTQ quantizer (all-MLP dCB = 0.615 vs. all-attention 0.083) and replicates across three families (Qwen 0.84, Llama 0.38, Gemma 0.36, all vs. attention ≤ 0.14); the localization holds in all five fact relations. We temper it honestly: the effect is weight-mass-confounded (MLP = 88% of parameters), and the equal-mass control that refutes the confound is *RTN-specific*, under GPTQ the mass-matched early-MLP effect vanishes (dCB = 0.000), so we claim *no* quantizer-general per-parameter localization; the early-MLP, mass-controlled refinement is a naive-RTN, Qwen-specific observation, and its depth is model-specific. The erasure is also graded, not surgical (MLP quantization raises perplexity 4.46× and halves MMLU), and rests on a single causal instrument: the MLP is best described as the *locus of the perturbation-induced recall collapse*, not the proven store of the fact.

We now draw out a deployment corollary. The component-level result of Sect. 4 carries a direct, if modest, engineering corollary. Quantizing the MLP erodes parametric recall far more than quantizing attention, so a deployment that supplies facts through retrieval no longer needs the MLP to be precise about *what the model knows*, it needs the MLP only to remain computationally intact, not accurate about stored facts. Attention, by contrast, is precision-robust, and the attention projections are comparatively small. The natural move is therefore a *knowledge-versus-context* precision allocation: starve the bulky MLP (which accounts for roughly 88% of the parameters in Qwen2.5-3B) of bits, and spend the saved budget on the comparatively cheap, precision-robust attention projections, where bits are inexpensive to protect. We stress at the outset that this is a corollary of the component-level mechanism, not an independent contribution: it is deliberately absent from the headline contributions of this paper, it applies *only* to pure-retrieval deployments (the allocation destroys closed-book memory, driving CB-Acc to 0.07), and we frame its gains conservatively throughout.

We compare four GPTQ [1] configurations against the same calibrated, strong baseline (anchor $n = 96$). Two are uniform (every projection at 3 or 4 bits); two are mixed, keeping attention at 8 bits while quantizing the MLP to 3 bits (average 3.6) or to 2 bits (average 2.7). Calibration, anchor set, and the metrics CB-Acc, CFR, OTHER, and open-book (support) accuracy are exactly as defined in Sect. 3. Results are reported in Table 14.

Table 14 GPTQ precision allocation (Qwen2.5-3B, $n = 96$); mixed holds attention at 8-bit and quantizes only the MLP. 95% Wilson CIs for CB-Acc/CFR/OTHER; MLP2/attn8 is non-functional (all 0)

Config	avg. bits	CB-Acc [95% CI]	CFR [95% CI]	OTHER [95% CI]	open-book
uniform-3	3.0	0.10 [0.06,0.18]	0.72 [0.62,0.80]	0.25 [0.17,0.35]	0.80
mixed MLP3/attn8	3.6	0.07 [0.04,0.14]	0.85 [0.77,0.91]	0.06 [0.03,0.13]	0.99
uniform-4	4.0	0.78 [0.69,0.85]	0.74 [0.64,0.82]	0.09 [0.05,0.17]	1.00
mixed MLP2/attn8	2.7	DEAD (0.00)			

The most tempting reading of Table 14 is the +0.19 jump in open-book accuracy from uniform-3 (0.80) to mixed-3.6 (0.99). We deliberately do *not* present this as a context-reallocation win, because the gain is almost entirely *garbage recovery*. Uniform-3 is partly broken, and its two garbage rates belong to different conditions: in the knowledge-conflict (closed-book) condition its OTHER rate is 0.25 (a quarter of responses are garbage), while in the support (open-book) condition its garbage stands at 0.20. Under mixed precision that support-condition garbage falls to 0.01. The open-book improvement (+0.19) is numerically indistinguishable from the garbage it eliminates (0.20 → 0.01). In other words, protecting attention does not teach the model to use context better here, it keeps the model *coherent per bit*. This is a functionality-per-bit story (Fig. 9b), and we present it as such.

The one comparison we are willing to defend is mixed-3.6 against *uniform-4*. At fewer average bits (3.6 versus 4.0), the mixed model matches uniform-4's functionality (OTHER 0.06 versus 0.09) while following context more often (CFR 0.85 versus 0.74). The improvement is real but modest, and it is decisively *not* a Pareto domination: open-book accuracy is tied at the ceiling (0.99 versus 1.00; McNemar $p = 1.0$), so the only edge is higher context-following at a lower bit budget. We therefore claim no more than that allocating precision toward attention can reach uniform-4 behaviour, with stronger context use, more cheaply.

This allocation is bought by destroying parametric memory: mixed-3.6 reads CB-Acc 0.07, lower even than uniform-4's 0.78. Such a model has essentially no usable closed-book knowledge and is appropriate *only* for pure-retrieval deployments where every fact arrives in context. The envelope is narrow on both sides. We restrict the recommendation to the *∼4-bit copy-RAG* regime, because at 3-bit the model is broadly degraded rather than surgically stripped of facts: the selectivity control (Sect. 4) shows that 3-bit MLP quantization raises perplexity by ∼4.5× and roughly halves MMLU, so the bits saved come with a general fluency and knowledge cost, not a free lunch. And the low end is a hard floor: pushing the MLP to 2 bits (mixed MLP2/attn8, 2.7 avg.) yields a dead model, consistent with the 3-bit MLP floor of Sect. 4, below it, the MLP's role as essential computation, not merely memory, dominates. Mixed precision thus buys a modest, RAG-specific edge around 4-bit, and we make that scope explicit rather than headline it.

In summary, allocating bits to precision-robust attention and starving the bulky MLP lets a mixed-3.6 GPTQ model match uniform-4's functionality (OTHER 0.06 vs 0.09)

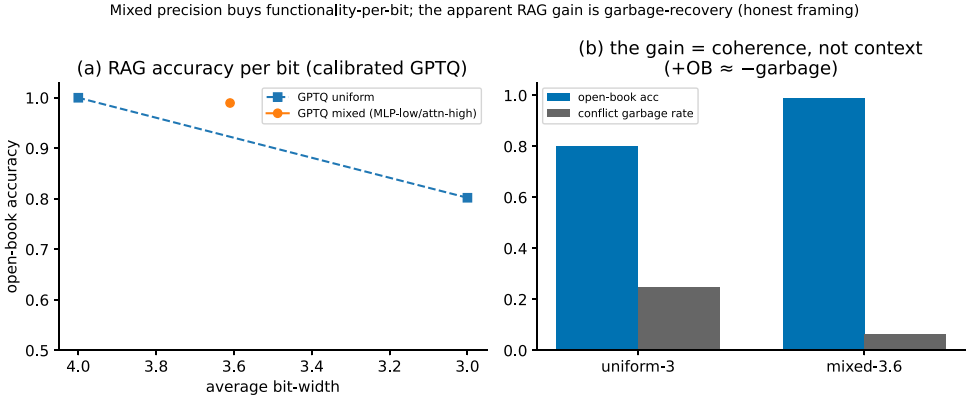


Fig. 9 Precision allocation under GPTQ (Qwen2.5-3B). (**a**) Open-book accuracy versus bit-width, uniform versus mixed. (**b**) The mixed-3.6 open-book gain is garbage recovery (support garbage **0.20** to **0.01**)

with higher context-following (CFR 0.85 vs 0.74) at fewer average bits, but the apparent open-book gain over uniform-3 is garbage recovery, not context reallocation, and the win is bought by destroying parametric memory (CB-Acc 0.07). We therefore narrow the recommendation to the ∼4-bit copy-RAG regime: at 3-bit the model is broadly degraded (PPL ∼4.5×, MMLU halved; Sect. 4), so this is a useful but modest, pure-retrieval-only corollary, not a free precision win.

We turn finally to deployment. The phenomenon of Sect. 4 establishes that low-bit PTQ erodes parametric recall while copy-from-context is the most robust capability. The deployment-relevant question is then *where* retrieval-augmented use survives quantization and where it does not. We answer with two retrieval tasks that bracket the deployment claim from both sides. The real result, the one that actually traces a degradation curve, is a deliberately hard, non-ceiling multi-hop task, which we lead with. We then bound the easy end with the single-hop copy task, whose recovery is at a ceiling and therefore informative only as an upper bound.

To obtain a genuine deployment curve where degradation is observable, we construct a harder retrieval task that forces *composition* rather than copying: a three-hop question over *fictional* entities (so nothing can be recalled parametrically), with the three gold facts placed non-adjacently among distractor sentences, requiring the model to chain them to an answer that appears *nowhere verbatim*. As in the comprehension probe of Sect. 4, the closed-book control is 0.000 for *every* configuration (n=80, full chain-of-thought scoring), so any open-book accuracy is purely contextual composition and not leakage. Unlike the single-hop copy task, this one is non-ceiling: even at full precision open-book accuracy is 0.912, leaving ample headroom to observe degradation.

Table 15 and Fig. 10 trace a real degradation curve. Open-book composition degrades *gracefully* through the moderate-precision regime, 0.912 at FP16, 0.850 at 8-bit, 0.812 at 4-bit, and then *collapses* at uniform 3-bit, falling to 0.013. Conflict-follow accuracy tracks the same trajectory (0.95 → 0.80 → 0.08). The collapse is not generic breakage but is MLP-driven, exactly as the component analysis predicts: at 3-bit, MLP-only quantization wrecks composition (0.138) while attention-only quantization largely preserves it (0.637). Multi-hop chaining is heavy MLP computation, so it follows the MLP below its operating floor; attention, being precision-robust, keeps most of the capability when only it is quantized.

The easy end of retrieval is, by contrast, a *ceiling*. In the support condition, each item presented with a single gold sentence stating the queried fact verbatim, open-book accuracy equals 1.00 across every bit-width of the retrieval-recovery sweep (`torchao` RTN, `bitsandbytes`, and GGUF k-quants, on Qwen2.5-3B and Qwen2.5-7B), down to and including 2-bit. Defining the recovery fraction

Table 15 Harder three-hop RAG (Qwen2.5-3B, fictional entities, n=80, CoT scoring; closed-book 0.000, 95% Wilson CI on fp16): degrades to 4-bit, collapses at uniform 3-bit; the failure is MLP-driven

Config	Bits	Closed-book	Open-book (compose)	Conflict-follow
FP16	16	0.000	0.912 [0.83,0.96]	0.95
8-bit	8	0.000	0.850	—
4-bit	4	0.000	0.812	0.80
3-bit (uniform)	3	0.000	0.013	0.08
MLP-only 3-bit	3	0.000	0.138	—
Attn-only 3-bit	3	0.000	0.637	—

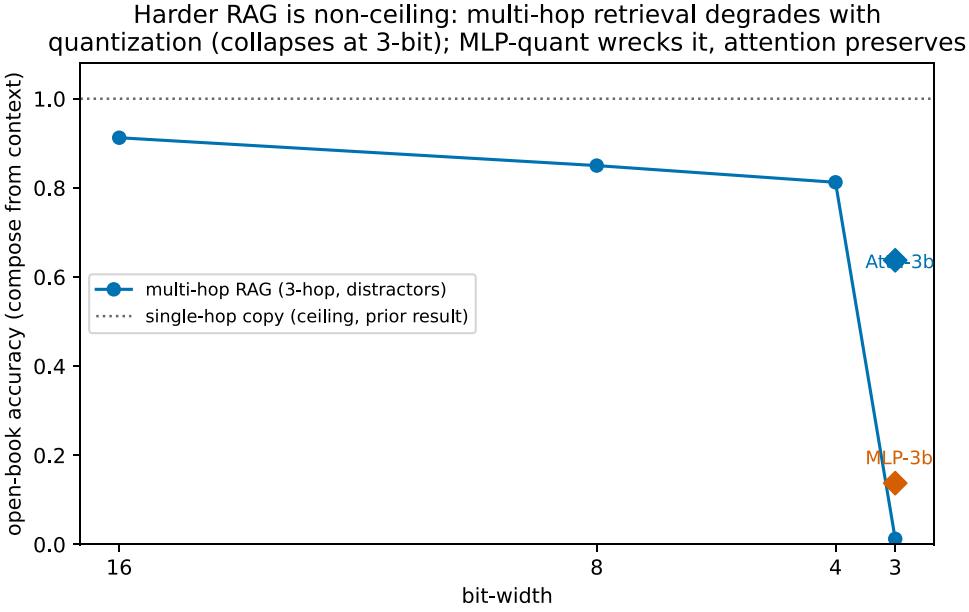


Fig. 10 Harder three-hop RAG (Qwen2.5-3B, fictional entities, closed-book = **0.000**): compose accuracy degrades **0.912** → **0.850** → **0.812** (fp16/8/4-bit), collapses at uniform 3-bit (**0.013**); 3-bit MLP-only **0.138**> attention-only **0.637**

Table 16 Single-hop recovery bound: Qwen2.5-3B (n=96) and Qwen2.5-7B (GGUF, n=100). CB-Acc drops but open-book holds at the **1.00** ceiling, so recovery is **100%** by construction; 95% Wilson CIs on CB-Acc

Config	Bits	CB-Acc [95% CI]	open-book	recovery
Qwen2.5-3B-Instruct (anchor n=96)				
FP16	16	1.00 [0.96,1.00]	1.00	n/a
8-bit	8	0.91 [0.83,0.95]	1.00	100%
4-bit	4	0.84 [0.76,0.90]	1.00	100%
Qwen2.5-7B-Instruct (anchor n=100)				
Q6	∼6	1.00 [0.96,1.00]	1.00	n/a
Q4	4	0.98 [0.93,0.99]	1.00	100%
Q3	3	0.95 [0.89,0.98]	1.00	100%
Q2	2	0.88 [0.80,0.93]	1.00	100%

$$\text{recovery} = \frac{\text{OB} - \text{CB-Acc}}{1 - \text{CB-Acc}} \tag{4}$$

as the share of the closed-book deficit restored once the fact is in context, OB = 1.00 forces recovery = 100%*by construction* at every bit-width (Table 16). We report this strictly as an upper bound, not as a claim of retrieval robustness: open-book accuracy is *constant*, not a graded curve, so a ceiling cannot reveal degradation. The task is deliberately easy, a single-sentence, distractor-free context in which the gold fact is a verbatim copy of the answer, so success reduces to copy-from-context. Within these limits the statement is clean: PTQ does not impair single-hop copy-from-context retrieval down to 2-bit (consistent with copy being the most robust capability of Sect. 4). This bound also scopes correctly: it is not a property of *every* config, the GPTQ uniform-3 do-or-die configuration of Sect. 4 reads open-book 0.80 because it is partly broken, and at the anchor/Q6 rows where CB=1.00 the fraction is 0/0 and undefined ("n/a").

The two retrieval results bound the deployment claim from both sides. Simple single-hop copy-from-context is *robust*: open-book accuracy stays at the 1.00 ceiling down to 2-bit, so a quantized model reliably answers when the fact is in the passage verbatim. Multi-hop reasoning over retrieved context is *not*: it degrades gracefully to 4-bit and collapses at uniform 3-bit, with the failure localized to the MLP. A practitioner can therefore quantize aggressively for copy-style RAG, but should treat 4-bit as a soft floor for retrieval that requires composition, and should not push uniform 3-bit where multi-hop chains are needed.

Table 17 consolidates these bounds into per-regime precision recommendations. The regimes differ in *which* capability they lean on, and the ordering of fragility established above dictates the safe floor for each: a closed-book, stored-knowledge deployment leans on the most fragile capability and should stay near 8-bit; single-hop copy-RAG leans on the most robust and tolerates very low bit-widths; multi-hop RAG leans on the precision-fragile MLP computation and must respect a 4-bit soft floor; and a retrieval-only, resource-constrained deployment can use the mixed MLP-3/attention-8 allocation of Sect. 4, accepting the destruction of closed-book memory that this entails.

Two production caveats bound these recommendations. First, the benefit of aggressive quantization is realized only to the extent that retrieval actually supplies the needed fact: our copy-RAG robustness assumes the gold fact is present in the passage, so in a real pipeline the quantized model's effective accuracy is bounded by the retriever's recall, a quantized reader cannot copy a fact its retriever failed to fetch, and closed-book fallback is exactly the capability that quantization has removed. Second, context length lowers the safe floor. The single-hop copy ceiling that holds to 2-bit in a short, distractor-free context drops to a 4-bit floor once the fact is buried among sixty distractors in a $\sim$570-word haystack (Table 18), and uniform 3-bit breaks long-context retrieval outright. Production deployments with long, distractor-heavy contexts should therefore adopt the more conservative long-context floors rather than the short-context ceilings.

In summary, the single-hop copy result is a ceiling (open-book = 1.00 to 2-bit, recovery 100% by construction); it bounds, but does not characterize, harder retrieval. On a non-ceiling three-hop composition task (closed-book = 0.000), open-book accuracy degrades gracefully (FP16 0.912 $\rightarrow$ 8-bit 0.850 $\rightarrow$ 4-bit 0.812) and then collapses at uniform 3-bit (0.013), with the failure MLP-driven (MLP-3b 0.138 vs attn-3b 0.637). Low-bit PTQ spares single-hop copy but degrades multi-hop reasoning over retrieved context.

The single-hop copy ceiling above is measured in a short, distractor-free context. A more demanding question is whether copy-from-context robustness, and the MLP locus,

Table 17 Deployment recommendations by regime, following the fragility ordering (recall most fragile, single-hop copy most robust). Floors are read from the Qwen2.5-3B/7B results of Sects. 4–4

Deployment regime	Safe precision floor	Rationale (from our results)
Closed-book / stored-knowledge inference	$\sim$8-bit; avoid $\leq$4-bit	Parametric recall is the most fragile capability; CB-Acc erodes first (Qwen-3B **1.00 $\rightarrow$ 0.84** at 4-bit; PopQA **1.00 $\rightarrow$ 0.71**).
Single-hop copy RAG (fact verbatim in context)	2-bit (short ctx); 4-bit (long ctx)	Copy-from-context is the most robust capability; open-book **= 1.00** to 2-bit short-context, and holds to 4-bit under needle-in-haystack long context.
Multi-hop / compositional RAG	4-bit soft floor; never uniform 3-bit	Composition is heavy MLP computation; degrades gracefully to 4-bit (**0.81**) then collapses at uniform 3-bit (**0.01**).
Retrieval-only, resource-constrained	mixed MLP-3/attn-8 ($\sim$3.6 avg. bits)	Matches uniform-4 functionality with higher context-following at fewer bits, but destroys closed-book memory (CB-Acc **0.07**), so it is appropriate only when every fact arrives in context.

Table 18 Long-context, retrieval-intensive ablations (Qwen2.5-3B, fictional entities, closed-book = 0.000): single-hop needle-in-a-haystack copy (n=120) and long multi-hop compose (n=100, CoT). Open-book holds through 4-bit then breaks at uniform 3-bit; MLP-only 3-bit destroys both while attention-only largely preserves copy. 95% Wilson CIs on open-book; long multi-hop is high-garbage (free-form CoT)

Config	Bits	open-book [95% CI]	conflict-follow	OTHER
Needle-in-a-haystack, single-hop copy (n=120)				
FP16	16	0.97 [0.92,0.99]	0.98	0.02
8-bit	8	0.94 [0.88,0.97]	0.94	0.06
4-bit	4	0.88 [0.80,0.92]	0.95	0.05
3-bit (uniform)	3	0.00 [0.00,0.03]	0.00	1.00
MLP-only 3-bit	3	0.01 [0.00,0.05]	0.06	0.94
Attn-only 3-bit	3	0.78 [0.70,0.85]	0.89	0.11
Long multi-hop compose (n=100, CoT)				
FP16	16	0.75 [0.66,0.82]	0.67	0.31
8-bit	8	0.64 [0.54,0.73]	0.55	0.42
4-bit	4	0.54 [0.44,0.63]	0.57	0.40
3-bit (uniform)	3	0.00 [0.00,0.04]	0.00	1.00
MLP-only 3-bit	3	0.03 [0.01,0.08]	0.05	0.95
Attn-only 3-bit	3	0.30 [0.22,0.40]	0.34	0.59

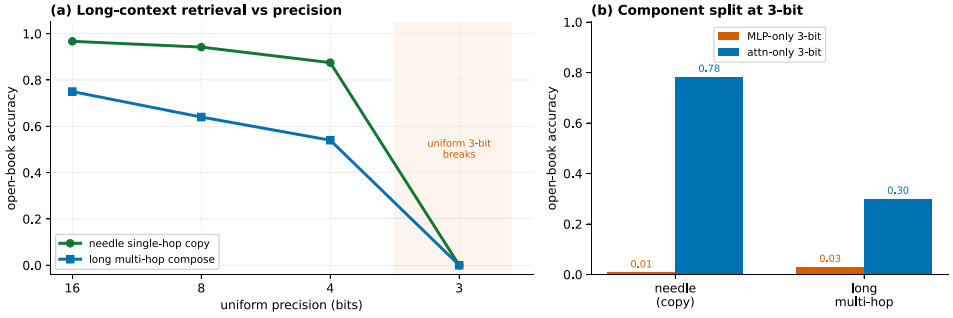


Fig. 11 Long-context, retrieval-intensive ablation (Qwen2.5-3B, fictional entities, closed-book 0.000): (**a**) open-book accuracy vs uniform precision for needle single-hop copy and long multi-hop compose, both breaking at uniform 3-bit; (**b**) At 3-bit, MLP-only quantization destroys both while attention-only largely preserves copy

survive under *long-context, retrieval-intensive* conditions. We therefore add two leak-free long-context probes over fictional entities (closed-book = 0.000 by construction, verified): a single-hop *needle-in-a-haystack* copy task, in which the gold fact is buried at depth among roughly sixty distractor company–city sentences in a ∼570-word context (n=120); and a *long multi-hop* compose task, in which a three-hop chain is scattered through a large haystack of ten distractor chains and filler (n=100, chain-of-thought scored). Table 18 and Fig. 11 report both. Even single-hop copy is no longer a flat ceiling under long context: open-book accuracy holds through 4-bit (0.97, 0.94,0.88 at FP16/8/4-bit) but breaks entirely at uniform 3-bit (0.00), and the component split is sharp, MLP-only 3-bit destroys long-context copy (0.01) while attention-only preserves it (0.78). Long multi-hop is non-ceiling even at full precision (compose 0.75) and degrades steadily (0.64, 0.54 at 8/4-bit) before collapsing at uniform 3-bit (0.00), with MLP-only 3-bit wrecking it (0.03) and attention-only partially preserving it (0.30); because free-form chain-of-thought yields a high garbage rate here (OTHER = 0.31 to 0.59), we read the compose-accuracy curve rather than the conflict rate. Both probes confirm the deployment story under long context: copy-from-context is the robust capability (now to 4-bit

rather than 2-bit, since long context is harder) and the precision-fragile retrieval computation lives in the MLP, while uniform 3-bit breaks long-context retrieval outright.

4.1 Discussion of numerical results

We consolidate here a reading of the principal numbers, so that the magnitudes in the preceding tables are interpreted rather than merely listed. Three quantities carry the argument: the closed-book accuracy CB-Acc (retained parametric memory), the context-following rate CFR on knowledge-conflict items, and the closed-book loss $\Delta\text{CB} = \text{CB-Acc}_{\text{FP}} - \text{CB-Acc}_q$, which is our effect-size measure for a quantization intervention.

The headline split is a divergence of two curves. On Qwen2.5-3B, as TORCHAO precision falls from FP16 to 4-bit, CB-Acc drops $1.00 \to 0.84$ while CFR rises $0.60 \to 0.74$; on Qwen2.5-7B the same opposition holds down to 2-bit (CB-Acc $1.00 \to 0.88$, CFR $0.74 \to 0.85$), with the garbage rate never exceeding 0.08. The importance of that last figure is easy to miss: because OTHER stays low, the two curves are separating because the model is changing *which* evidence source it trusts, not because it is breaking. The rising CFR must not be read as improved context use. On the component-wise MLP-3-bit configuration, where the rise is largest ($0.604 \to 0.823$), about 96% of the newly context-following items (25 of 26 flips) are precisely those that also lost their closed-book memory, and on the subset whose memory survives, CFR barely moves ($0.667 \to 0.733$). The number to trust for context ability is therefore the leak-free comprehension probe, whose closed-book accuracy is 0.00 by construction: genuine two-hop composition survives 8-bit (0.97) and largely 4-bit (0.74) before collapsing at uniform 3-bit (0.15).

The component numbers localize where the fragile recall lives. At 3-bit on Qwen, quantizing only the MLP erases $\Delta\text{CB} = 0.84$ of closed-book memory while quantizing only attention erases 0.14: a roughly six-fold gap whose 95% intervals do not overlap, and which reproduces across families (Llama 0.38 vs. 0.03; Gemma 0.36 vs. 0.02) and survives a calibrated quantizer (GPTQ 0.62 vs. 0.08). Calibration shrinks the absolute erasure, as one expects from a smarter quantizer, but does not move the ordering. The one place calibration *does* overturn a result is the sharper equal-mass claim: at a matched $\sim$340M perturbed weights, the early MLP layers erase $\Delta\text{CB} = 0.667$ under naive round-to-nearest but 0.000 under GPTQ, which is why we scope the early-layer localization to round-to-nearest and Qwen and claim only the component-level result across quantizers.

Finally, the erasure is graded, not surgical, and the numbers say so plainly: the MLP-3-bit model that loses 71% of its recall ($\Delta\text{recall} = 0.711$) also raises WikiText-2 perplexity by 4.46× and roughly halves MMLU ($0.657 \to 0.306$), while its verbatim copy-from-context accuracy stays at 1.00. The finding is thus an *ordering* of fragility, recall most fragile, fluency and multiple-choice reasoning intermediate, copy-from-context most robust, rather than a claim that facts are excised while the rest of the model is untouched. This ordering is what licenses the deployment reading of Sect. 4: closed-book benchmarks probe exactly the capability quantization destroys first, and so overstate its harm for retrieval-augmented use whose reliance is on reading supplied context.

5 Limitations and threats to validity

We state PRISM's limitations candidly, bounding each claim to the regime in which it holds and reporting garbage (OTHER) rates throughout rather than hiding breakage behind a diluted CFR.

5.1 Scale

All experiments run on a single 8 GB consumer GPU (RTX 5060), inference-only, restricting us to models $\leq$ 7B. The component-wise mechanism and equal-mass control are characterized at 3B (Qwen), because component-wise quantization needs an FP16 base resident alongside its quantized variant and a 7B FP16 model alone is $\sim$15 GB. We therefore scope deployment recommendations to $\leq$ 7B, argue the phenomenon's scaling only qualitatively, and leave confirming the mechanism at 13–70B to larger-GPU work. A second scope caveat concerns our reliance on synthetic probes. The in-house conflict and comprehension sets use synthetic and fictional-entity constructions *deliberately*, because only invented entities can guarantee a leak-free measurement (closed-book accuracy 0.00 by construction), which no naturally occurring benchmark can provide. To offset the external-validity cost of that choice, the same phenomenon and the same MLP localization are also demonstrated on two standard public factual benchmarks (PopQA and CounterFact) and against two standard capability suites (MMLU and WikiText-2), so the conclusions do not rest on synthetic data alone. Broader validation on naturally sourced, real-retriever corpora nonetheless remains future work.

5.2 Localization specificity

Our sharpest claim, that at *equal perturbed mass* quantizing the *early* MLP layers of Qwen2.5-3B erases more recall (early-5 MLP ΔCB = 0.667) than all of attention (0.135) or the mid-5 MLP (0.010), holds only under naive round-to-nearest: under calibrated GPTQ the effect vanishes (ΔCB = 0.000) and attention then erodes more, so the mass-controlled refutation of the weight-mass confound is round-to-nearest-specific. What survives a change of quantizer is the weaker, *weight-mass-confounded* component-level contrast (MLP vs. attention ΔCB = 0.84 vs. 0.14 RTN, 0.62 vs. 0.08 GPTQ); since the MLP holds $\sim$88% of parameters with comparable per-parameter sensitivities, we do *not* claim a quantizer-general per-parameter localization. Depth is likewise not general (Gemma-2-2B early-5 MLP ΔCB = 0.019), and mass cannot be matched for early attention ($\sim$47 M vs. $\sim$340 M params), so a layer-matched control (ΔCB = 0.677 vs. 0.021 at the first six layers) leaves depth and component partly entangled. We therefore present early-MLP as a round-to-nearest, Qwen-specific observation and claim cross-quantizer/cross-family only the component-level result (Llama ΔCB = 0.38/0.03, Gemma 0.36/0.02). The phenomenon is also not universal: Llama-3.2-3B shows *no* closed-book erosion under 4-bit `bitsandbytes` (1.00 $\rightarrow$ 1.00), which we report rather than suppress.

5.3 Selectivity of knowledge degradation

Quantization does not erase facts while leaving the model intact: at 3-bit, MLP quantization also raises WikiText-2 perplexity $\sim$4.5$\times$ and roughly halves MMLU. The honest claim is an *ordering*, recall most fragile (Δrecall = 0.711 at MLP-3-bit), verbatim copy-from-context most robust (support 1.00), reasoning and fluency between, so we narrow deployment recommendations to $\sim$4-bit copy-RAG. The apparent rise in

counterfactual-following under MLP quantization ($0.60 \rightarrow 0.82$) is largely removed competition: $\sim$96% of the additional context-following items (25 of 26 flips) also lost their closed-book memory, and on the memory-intact subset the rate barely moves ($0.667 \rightarrow 0.733$); we therefore lead with the leak-free comprehension probe.

5.4 Evidence quality

Our mechanistic claims rest on one causal instrument, component-wise quantization, which shows the MLP *matters* for recall under perturbation, weaker than "the MLP *stores* the fact"; converging evidence (causal tracing, editing at matched precision) is future work. A second instrument, activation patching ($n = 60$), corroborates only weakly, being confounded by distribution mismatch (it restores memory non-selectively, MLP $0.24 \rightarrow 0.57$, attention $\rightarrow 0.63$), so we treat it as an appendix note. External corroboration on CounterFact [13] replicates the direction but with high cloze garbage (OTHER ≈ 0.25); we report it as supportive-but-noisy.

A further confound is intrinsic to the component-wise instrument and we cannot fully resolve it. Reducing precision in one layer alters the residual distribution flowing to subsequent layers, so quantizing the MLP changes not only that sublayer's numerical fidelity but the dynamic range and noise floor of everything computed downstream of it. Our finding that MLPs are quantization-sensitive could therefore reflect MLPs' intrinsic precision requirements, the cascading effects of altered residuals on downstream layers, or a combination of both; the present design cannot separate these. Isolating them would require a different design, layer-wise interventions that reduce precision while preserving residual statistics, or selective weight quantization paired with layer-norm adjustments, which we leave to future work. We note that this confound tempers the mechanistic *interpretation* but not the deployment-facing observation itself: whatever its ultimate cause, quantizing the MLP is what erodes recall, and this is exactly what the precision-allocation corollary exploits.

We also note, in the interest of foregrounding experimental rigor, that the central component claim is not carried by a single comparison but by a battery of adversarial controls, each designed to break it: an equal-perturbed-mass control ($\sim$340 M weights matched across early-MLP, attention, and middle-MLP), a cross-quantizer control (naive round-to-nearest versus calibrated GPTQ, which *refutes* our own sharper early-MLP claim and which we report against interest), Control C (a harder 2-bit budget on attention still spares more memory than a milder 3-bit budget on the MLP), Control C′ (matched *measured* dequantization error rather than bit-width or mass), a selectivity control (MMLU and perplexity, establishing the erasure is graded not surgical), anchored-versus-unfiltered robustness, a seed-variance check, and two-annotator validation of the classifier. The claims that survive all of these are the ones we headline; those that do not (the mass-controlled early-MLP localization under GPTQ) we explicitly scope down.

5.5 Task scope

Our retrieval-robustness claim is specifically single-hop *verbatim copy*: with the gold fact placed verbatim, open-book accuracy is 1.00 down to 2-bit, a ceiling that does *not* extend to multi-hop. The leak-free comprehension probe (closed-book 0.00) survives 8-bit and largely 4-bit but breaks at uniform 3-bit (MLP-only 0.43 vs. attention-only 0.94), and the

harder 3-hop task is non-ceiling at full precision (0.91), degrading to 0.81 at 4-bit before collapsing. Anchoring on items FP16 answers closed-book forces baseline CB-Acc to 1.00 by construction; unfiltered numbers barely move (MLP-3 CFR 0.823 anchored vs. 0.807). Finally, uniform quantization at $\leq$ 3-bit can break the model outright, so we flag any OTHER > 0.2 configuration as a breakage regime and restrict claims to the functional regime.

6 Conclusion

Using PRISM, we have shown that post-training quantization does not degrade a language model uniformly; it fragments capabilities by their *source*. Open-ended *parametric recall* collapses under low-bit PTQ while verbatim *copy-from-context* is most robust and multi-hop comprehension is intermediate: on Qwen2.5-3B/7B closed-book accuracy falls from 1.00 to 0.84 (3B, 4-bit) and 0.88 (7B, 2-bit) with garbage below 0.08, while copy-from-context stays at ceiling and leak-free comprehension survives to 4-bit. Component-wise, quantizing only the MLP collapses recall while quantizing only attention spares it (round-to-nearest ΔCB = 0.84 vs. 0.14), a contrast that survives a change of quantizer and replicates across three families, relating the MLP-as-memory view [12, 13, 18] to the precision axis. We do not over-read it: the contrast is *weight-mass-confounded* ($\sim$88% of parameters in the MLP), the sharper early-MLP localization is specific to round-to-nearest on Qwen and removed by calibrated GPTQ, and we therefore describe the MLP as the *locus of the perturbation-induced recall collapse* rather than where the fact is stored.

We are honest about scope. The erasure is *graded, not surgical* (at 3-bit, MLP quantization raises perplexity $\sim$4.5$\times$ and halves MMLU), the apparent rise in counterfactual-following is largely *removed competition* (evidenced instead by the leak-free probe), and the remaining limits, a single 8 GB GPU bounding models to $\leq$7B and causal evidence from one instrument, are detailed in Sect. 5. The deployment takeaway: closed-book benchmarks *overstate* the harm of quantization for single-hop copy-RAG, which stays at ceiling to 2-bit, but multi-hop retrieval is MLP-fragile and *not* spared.

Future work should test whether the fragmentation and component-level localization persist at 13–70B, in long-context settings with many distractors, across languages beyond the English probes used here (whether the memory-versus-context split and the MLP locus hold multilingually is open), and under distractor-heavy real-retriever RAG. A natural mechanistic question is *why* a calibrated quantizer reshapes the within-MLP localization that round-to-nearest produces, bearing on quantizer-aware methods that protect parameters carrying parametric recall. The broader hope is a precision budget assigned by *what a deployment needs to remember versus read*; we release PRISM, with its probes, data, and logs, to support that.

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Author contributions
S.Vaish and A.Gautam were involved in the ideation, planning, experimentation, coding, and writing the initial draft. V.Tiwari and J.Yadav were involved in ideation, Analysis, supervision, and preparing the final draft.

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Data availability
The datasets used in this article are properly cited and referenced within the manuscript. The dataset utilized for experimental analysis is publicly available and can be accessed at: CounterFact-Tracing (Hugging Face identifier: NeelNanda/counterfact-tracing) https://huggingface.co/datasets/NeelNanda/counterfact-tracing PopQA: A Large-Scale Open-Domain Question Answering Dataset: https://huggingface.co/datasets/akariasai/PopQA MMLU (Massive Multitask Language Understanding): https://huggingface.co/datasets/cais/mmluWikiText (Hugging Face identifier: Salesforce/wiki text)): https://huggingface.co/datasets/Salesforce/wikitext.

Code availability
The full experimental pipeline, dataset generators, component-wise and mixed-precision quantizers, scoring, the negation-aware classifier with its human-label validation, and the confidence-interval and figure scripts, together with the derived probe data and per-item result files backing every reported number, is publicly available at https://github.com/Swa-s-tik/PRISM (an archived release with a citable DOI will be deposited on Zenodo upon acceptance).

Declarations

Ethics approval
Not applicable.

Consent to participate
Not applicable.

Consent to publish
Not applicable.

Competing interests
The authors declare no competing interests.

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